Refinement and Inference Growth in SKJS: a Soundness Model of the Type Checker

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Abstract

We model SKJS's internal type representation and its two flow-sensitive type transformations: predicate-driven *refinement* (control-flow narrowing in `evalCond`) and *constant removal* (the remaining use of the checker's former general type-difference operation). We prove soundness properties for both transformations and model the inference substitution as a state machine, making explicit the conditions under which inference growth (`growTVars`) preserves membership in types computed earlier during refinement. This is not a claim that the whole checker is sound. Every implementation behavior needed by the results is either modeled here, represented by a named registry assumption or obligation, or excluded with a reason that it cannot affect the stated result. The registry and exclusions appear in Sections 8 and 1. This document supersedes the draft attached to PR #342 ("Refinement and Type Difference in SKJS").

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Reading guide. Sections 1–6 build the model and prove the refinement, constant-removal, and inference results. The worked examples then show both established behavior and known counterexamples. Readers auditing a particular assumption can go directly to Section 8, whose generated registry entries name the owning evidence paper and implementation anchors. The final two sections record implementation correspondence and the maintenance rules for keeping the corpus synchronized with the checker.

1 Scope, Method, and Modeling Contract

Pinned revision. All implementation references (file, class, function) are against commit `bafdcf7c`. The document is maintained by the discipline of Appendix B: every inference rule in the refinement master list corresponds to one dispatch arm of `evalCond`, and drift between the list and the dispatch is intended to be mechanically checkable. Two terms of art for that mechanism, used throughout: an *implementation anchor* is a file/class/function reference (typeset as impl: `...`) locating the code a definition or rule models — the maintenance check verifies anchors bidirectionally; a *pinned test* is a test in the repository that fails when a documented behavior changes, forcing the corresponding passage to be updated in the same change.

Modeling contract. Every behavior of the implementation that can affect type soundness is placed in exactly one of three bins:

1. **Modeled**: defined concretely in this document and proved sound.
2. **Abstracted**: hidden behind an interface with explicitly stated axioms. All such axioms are collected in Section 8 (Assumptions and Proof Obligations), each with an implementation anchor and a status from the fixed vocabulary PROVED / ASSUMED / OPEN / VIOLATED defined there.
3. **Out of scope**: listed below, each with a reason why omission cannot create a hole in the soundness argument.

An implementation behavior that fits none of the bins is a defect of this document.

Terminology. The model deliberately keeps the implementation's names so that every term can be grepped for; the cost is a few SKJS-specific usages, fixed here. An *object branch* is one alternative of an object type's internal union (the implementation's `TObject.branches`); it is always qualified, since *true/false branch* refers to control flow. A *member* is one SType alternative of an STypes union; a *clause* pairs type parameters with such a union. A *candidate* is an ordinary inference variable (`TVID`); *grow* and *seal* are the substitution transitions of Section 6; *expansion* rewrites a type to its instance-plus-clauses view under the current substitution. A *brand* is a type-level claim about a runtime property that refinement treats as evidence. The array brand records the result of an `Array.isArray` test on an object branch; CAbsent is a brand-like claim that the branch contains no callable values. a *phantom field* records runtime-confirmed presence without a declaration. "Open" is used only for the substitution state SVOpen and the obligation status; the object semantics of Definition 13 is called *extensible*.

Routing source features into the contract. Classification follows implementation mechanisms rather than whole source-language features. A single feature can have a modeled type representation, a checking phase represented by a registry obligation, and a diagnostic outside this theorem.

The following normalization results are modeled directly because they produce the literal and object forms defined in Section 3: finite template literal types become finite unions of string literals; `enum` types become unions of their member literal types; and `Closed<T>` becomes `T`'s object type with never-valued indexes for uncovered keys. The registry records the required properties of the code that produces these forms:

- **L-LIT** covers the representation of every stored literal, whether produced by the parser or the enum constant folder.
- **L-LIT-FOLD** separately requires an enum member's synthesized literal type to denote the member's runtime value.
- **L-EXP** covers expansion of the normalized types.
- **L-SUB-CLOSED-ABSENCE** governs when subtyping may establish the absence facts consumed by rule R-In.

A `switch` case uses the modeled R-Eq rules, and fallthrough adds the extra scope entry shown in the refinement-rule list below. Whether a finite set of cases covers every value is instead a statement-level property for the future preservation theorem. Such a result will require **L-LIT-FOLD**, because coverage depends on the runtime values of enum members.

Other mechanisms remain represented by registry rows or lie outside the one-step result. Expression typing is **L-TC**. Behavioral function signatures are E-SIG, which covers synthesized `.call`/`.apply`/`.bind` members, explicit `this`-parameter checking, and contextual typing of unannotated closure parameters. Executing an enum declaration and constructing its runtime forward/reverse-mapping object belongs to the future statement-preservation theorem. This exclusion does not hide the enum object's type-level components from the modeled results: refinement and subtyping still see its ordinary fields and callability mode. Switch-exhaustiveness reporting is a nonfatal diagnostic and does not affect acceptance. Each mechanism is therefore classified where a proof uses it; no source-language feature receives one blanket classification.

impl: `TEnv.sk / normalizeTemplate, closeObjectType; TypeCheckEnum.sk; TypeCheckStmt.sk / Switch; TypeCheckExpr.sk / call checking`

Out of scope, with justification.

- *Syntax rejected at normalization.* Conditional types (`T extends U ? X : Y`, SKJS9020), `infer` (9017), indexed access `T[K]` (9021), type predicates `x is T` (9015), `asserts` predicates (9013), and template literal types with an infinitely-valued hole (9018): these produce a fatal diagnostic in `TEnv::normalize` and never reach the type machinery, so the soundness claim does not cover them.

  impl: `TEnv.sk / normalize`

- *Unhandled condition forms.* The refinement dispatch falls through to the identity on any condition shape it does not recognize (Rule R-Other), and the identity refinement is trivially sound (Lemma 24). Consequently a stale or incomplete master list makes this document *incomplete*, never *unsound*, as a description of narrowing.

- *Mutation during checking of straight-line code.* This document proves *one-step branch soundness*: a single condition is evaluated against a fixed store. Assignment, loops, and join points are treated only through the union-soundness of scope merging (Lemma 35); a full small-step preservation theorem is future work. The interaction of narrowing with mutation *during condition evaluation* is governed by assumptions **L-PURE** and **L-LOCAL**.

- *Flow-typing sites outside condition narrowing.* The master list covers condition forms only. The checker performs flow-sensitive type updates at five further sites, listed here so the boundary is a stated enumeration rather than an implication: assignment to a local updates its scope entry (`TypeCheckExpr.sk`, `Var`-LHS `scope.set`); literal-typed local objects grow fields on assignment; `this` is re-typed after `super(...)`; the non-null assertion `x!` and optional chaining remove nullish members from the *expression's* type (no scope mutation, via `removeUndefined`); and the `for-in` index mechanism suppresses element errors without narrowing. None of these is a condition refinement; each would be a case of the future preservation theorem. Note also that `for`-loop conditions do *not* narrow.

- *Numeric coercions, operators, and standard-library semantics.* Only the condition forms of the master list are given evaluation semantics.

2 Runtime Model

Definition 1 (Runtime Tags)**.** Tags *is the nine-element set*

$$\mathsf{Tags} = \{\mathsf{undef}, \mathsf{void}, \mathsf{null}, \mathsf{bool}, \mathsf{num}, \mathsf{str}, \mathsf{bigint}, \mathsf{sym}, \mathsf{obj}\}.$$

impl: `TType.sk / STypeTag (TagUndefined ... TagObject)`

There is no dedicated tag for tuples, promises, or algebraic data types: tuples and promises are object-tagged values whose types live in the instance layer or as object types (`SKJS_Tuple`/`ReadonlyTuple` instances; `Promise` as a stub-declared object type), and discriminated unions are ordinary object types with discriminant fields.

Definition 2 (Runtime Values)**.** *Fix payload sets* $(\mathsf{Payload}_k)_{k\in\mathsf{Tags}}$ *and define* $\mathsf{Val} \triangleq \{(k,p) \mid k \in \mathsf{Tags},\ p \in \mathsf{Payload}_k\}$*, with* $\mathsf{rtag}((k,p)) = k$*. Primitive payloads are concrete; object payloads are abstract:*

$$\mathsf{Payload}_{\mathsf{undef}} = \{\star\} \qquad \mathsf{Payload}_{\mathsf{void}} = \{\star\} \qquad \mathsf{Payload}_{\mathsf{null}} = \{\star\}$$
$$\mathsf{Payload}_{\mathsf{bool}} = \{\mathsf{true},\mathsf{false}\} \qquad \mathsf{Payload}_{\mathsf{num}} = \mathbb{F} \qquad \mathsf{Payload}_{\mathsf{str}} = \mathsf{String}$$
$$\mathsf{Payload}_{\mathsf{bigint}} = \mathbb{Z} \qquad \mathsf{Payload}_{\mathsf{sym}} = \mathsf{SymVal} \qquad \mathsf{Payload}_{\mathsf{obj}} = \mathsf{Payload}'_{\mathsf{obj}}$$

where $\mathbb{F}$ *is the set of IEEE-754 doubles (including* -0 *and* NaN*),* SymVal *an abstract infinite set of symbol identities, and* $\mathsf{Payload}'_{\mathsf{obj}}$ *an arbitrary set. The tag* void *models the checker's distinct* `void` *type; at runtime its unique payload is the value* `undefined`*, which is why refinements treat the tags* undef *and* void *together (Definition 26).*

Definition 3 (Object Interface)**.** *Let* $\mathsf{ObjVal} \triangleq \{v \in \mathsf{Val} \mid \mathsf{rtag}(v) = \mathsf{obj}\}$*. We assume the following abstract interface on object values:*

- *Keys:* $\mathsf{Key} \triangleq \mathsf{String} \uplus \mathsf{SymVal}$*.*
- $\mathsf{props} : \mathsf{ObjVal} \to \mathcal{P}(\mathsf{Key})$ *and a partial* $\mathsf{get} : \mathsf{ObjVal} \times \mathsf{Key} \rightharpoonup \mathsf{Val}$*, with* $\mathsf{get}(o,p)$ *defined iff* $p \in \mathsf{props}(o)$*. Write* $\mathsf{has}(o,p)$ *for* $p \in \mathsf{props}(o)$*.*
- $\mathsf{callable}, \mathsf{constructible} : \mathsf{ObjVal} \to \{\mathsf{true},\mathsf{false}\}$*, with the JS fact* $\mathsf{constructible}(o) \implies \mathsf{callable}(o)$ *(every constructor is a function;* `typeof` *yields* `"function"` *for it).*
- $\mathsf{isArrayV} : \mathsf{ObjVal} \to \{\mathsf{true},\mathsf{false}\}$*, the runtime truth of* `Array.isArray`*.*
- *A classification* $\mathsf{kind} : \mathsf{String} \to \{\mathsf{strKey},\mathsf{numKey}\}$ *of string keys into numeric and non-numeric spellings, used by index signatures.*

This interface is a snapshot view: get *describes the properties of* o *at the moment a condition is evaluated. See assumption* ***L-PURE****.*

Definition 4 (Strict Equality)**.** *JS strict equality is* not *identity on* Val*; we model it as a relation* $\approx \subseteq \mathsf{Val} \times \mathsf{Val}$*:* $v \approx w$ *iff one of*

- $\mathsf{rtag}(v), \mathsf{rtag}(w) \in \{\mathsf{undef},\mathsf{void}\}$ *(the* `void` *tag is a checker fiction for the runtime value* `undefined`*);*
- $\mathsf{rtag}(v) = \mathsf{rtag}(w) = \mathsf{num}$ *and the payloads are equal as IEEE doubles with* -0 *identified with* 0 *and* NaN *equal to nothing (including itself);*
- $\mathsf{rtag}(v) = \mathsf{rtag}(w) \notin \{\mathsf{undef},\mathsf{void},\mathsf{num}\}$ *and the payloads are equal (reference identity for objects).*

Note $\approx$ *is not reflexive (*$\mathsf{NaN} \not\approx \mathsf{NaN}$*) and crosses the* undef/void *tag pair. A set* $X \subseteq \mathsf{Val}$ *is* $\approx$-closed *if* $v \in X$ *and* $w \approx v$ *imply* $w \in X$*.*

Lemma 5 ($\approx$ is a Partial Equivalence)**.** $\approx$ *is symmetric and transitive; its domain (the values related to themselves) is* Val *minus the* NaN*-payload numbers. On that domain "the* $\approx$*-class of* v*" is well-defined.*

Proof. Each arm is symmetric. Transitivity: a chain stays within the $\{\mathsf{undef},\mathsf{void}\}$ pair or within a single tag with payload equality (for numbers, equality-with-$\pm$0-identified is transitive and NaN relates to nothing, so chains through it are vacuous). □

Definition 6 (Truthiness)**.** $\mathsf{truthy} : \mathsf{Val} \to \{\mathsf{true}, \mathsf{false}\}$ *is defined per tag: values of tags* undef, void, null *are falsy;* (bool, b) *is truthy iff* $b = \mathsf{true}$*;* (num, n) *is falsy iff* $n \in \{0, -0, \mathsf{NaN}\}$*;* (str, s) *is falsy iff* $s =$ `""`*;* (bigint, z) *is falsy iff* $z = 0$*; symbols and objects are always truthy.*
impl: `TType.sk / TPrimValueKind::isFalsy, SType::falsiness`

3 Internal Type Representation

3.1 Simple types

Definition 7 (Simple Types)**.** SType *has exactly three constructor families:*

- $\mathsf{unit}(u)$ *for* $u \in \{$`undefined`, `void`, `null`$\}$*;*
- $\mathsf{prim}(k, L)$ *for* $k \in \{$`bool`, `num`, `str`, `bigint`, `sym`$\}$ *and* $L \in \{\top\} \cup \mathcal{P}^{+}_{\mathrm{fin}}(\mathsf{Lit}_k)$*, where* Lit_k *is the set of literal spellings of kind* k *and* $\top$ *means the full primitive;*
- $\mathsf{object}(B)$ *for a nonempty finite set* B *of* branches *(Definition 11).*

tag *maps each constructor to the evident tag.*
impl: `TType.sk / TPrimUnit, TPrimValue, TObject; SType::tag`

Definition 8 (Literal Denotation)**.** *Assume a denotation* $\delta_k : \mathsf{Lit}_k \to \mathsf{Payload}_{\mathsf{tag}(k)}$ *from literal spellings to payloads, and define*

$$\begin{aligned}
[\![\mathsf{unit}(u)]\!] &\triangleq \{(\mathsf{tag}(u), \star)\} \\
[\![\mathsf{prim}(k, \top)]\!] &\triangleq \{(\mathsf{tag}(k), p) \mid p \in \mathsf{Payload}_{\mathsf{tag}(k)}\} \\
[\![\mathsf{prim}(k, L)]\!] &\triangleq \{\, v \mid \mathsf{rtag}(v) = \mathsf{tag}(k),\ \exists \ell \in L.\ v \approx (\mathsf{tag}(k), \delta_k(\ell)) \,\}.
\end{aligned}$$

Literal-type denotations are the same-tag $\approx$-closure *of their spellings' denotations: the literal type* `0` *contains both* 0 *and* -0*, matching JS strict equality and tsc's treatment of numeric literal types. (For non-numeric kinds the closure is the identity. The cross-tag* $\mathsf{undef}/\mathsf{void}$ *arm of* $\approx$ *is deliberately* not *included in unit denotations; the rules that compare against* `undefined` *handle the two tags jointly, and the residual corner is assumption* ***L-SEQ****.) We assume* Lit_{num} *contains no* NaN *spelling: there is no NaN literal type, and a hypothetical one would denote* $\emptyset$ *under closure semantics (nothing is* $\approx \mathsf{NaN}$*), making* ***L-TC*** *unsatisfiable for it. The implementation stores literal* spellings *and compares them syntactically; assumption* ***L-LIT*** *states the assumptions this requires. Unique symbols are represented as literals of kind* `sym` *whose stored string is a* synthesized position-derived identity *(`unique_symbol:file:sym:offset:name`), not a source lexeme — so distinct unique symbols have distinct denotations by construction.*
impl: `PrimLiterals.sk; TType.sk / TPrimValue; TEnv.sk / normalize (TAUniqueSymbol)`

Lemma 9 (A0: Tag Consistency)**.** $\forall s \in \mathsf{SType},\ v \in [\![s]\!].\ \mathsf{rtag}(v) = \mathsf{tag}(s)$.

Proof. Immediate from the definitions above and Definition 13, whose members are drawn from ObjVal. □

Lemma 10 (A1: Tag Disjointness)**.** $\forall s_1, s_2 \in \mathsf{SType}.\ \mathsf{tag}(s_1) \neq \mathsf{tag}(s_2) \implies [\![s_1]\!] \cap [\![s_2]\!] = \emptyset$.

Proof. By A0, as values carry a unique tag. □

3.2 Object branches and fields

Definition 11 (Fields and Branches)**.** *A* field *is a record* $f = (\mathsf{name}, \mathsf{opt}, \mathsf{get}, \mathsf{set}, \mathsf{access})$ *with* $\mathsf{name} \in \mathsf{Key}$*,* $\mathsf{opt} \in \{\mathsf{true}, \mathsf{false}\}$*,* $\mathsf{get}, \mathsf{set} \in \mathsf{TType}$*, and* $\mathsf{access} \in \{\mathsf{Pub}, \mathsf{Priv}(\mathit{owner}), \mathsf{Never}, \mathsf{Phantom}\}$*. An* object branch $b \in \mathsf{TFields}$ *is a record consisting of:*

- *finite maps* $\mathsf{flds}, \mathsf{symflds}$ *from string/symbol keys to fields;*
- *optional index fields* idx_κ *for* $\kappa \in \{\mathsf{strKey}, \mathsf{numKey}, \mathsf{symKey}\}$*, each a field;*
- $\mathsf{call}, \mathsf{new} \in \mathsf{Callability}$*, the four-point lattice of Definition 12, whose* CPresent *point carries an abstract function signature* σ*;*
- $\mathsf{isArr} \in \{\mathsf{None}, \mathsf{Some}(\mathsf{true}), \mathsf{Some}(\mathsf{false})\}$*, the array brand.*

Readonly fields are encoded as $\mathsf{set} = \bot$ *(`TField::create` with `isReadonly`). The implementation's `TObject` additionally carries `unexpandedType : ?TSInst`, a best-effort record of the named type an object came from; it is a printing hint plus one program-*restricting *check (the async-return "must be global `Promise`" rule) and never participates in* $[\![\cdot]\!]$*, subtyping, or refinement, so the model omits it.*

impl: `TType.sk / TField, TFields, TIndexesBase, FieldAccess`

Definition 12 (Callability)**.** Callability *is the four-point lattice*

$$\mathsf{CPresent}(\sigma) \;\leq\; \mathsf{CFunction} \;\leq\; \mathsf{CUnknown}, \qquad \mathsf{CAbsent} \;\leq\; \mathsf{CUnknown},$$

read as a constraint on the members of an object branch — for call *on* [[Call]]*, for* new *on* [[Construct]]*:* CPresent(σ)*, every member is invocable with signature* σ*;* CFunction*, every member implements the internal method but no signature is known, so it is* not *invocable (the denotation of the global `Function` type — invocation requires* CPresent*);* CAbsent*, no member implements it;* CUnknown*, no constraint — the mode of structural object types ({}, interfaces, object annotations), whose width subtyping admits callable inhabitants (`const o: {} = () => {}`). Two invariants hold at branch construction: an array-branded branch lowers* CUnknown *to* CAbsent *(arrays are never callable, provided the array brand is accurate), and a constructible branch (*new $\in \{\mathsf{CPresent}, \mathsf{CFunction}\}$*) raises an unknown* call *to* CFunction*, by the* constructible $\implies$ callable *fact of Definition 3 — which is what lets* R-TYPEOF *decide typeof-`"function"`-ness from the* call *mode alone. A class-constructor* value *is typed* new = CPresent(σ_C)*,* call = CFunction *(its* [[Call]] *always throws, so there is no signature to offer) — registry row A-CTOR-CALL. Nominal class instances and enum objects carry* call = new = CAbsent*, a brand-like approximation that structural width subtyping cannot fully enforce — registry row F-CABSENT records whether refinement may rely on that claim; interfaces carry* CUnknown *for both.*

impl: `TType.sk / Callability, TFields::createFromSortedIterator; TEnv.sk / newStaticObject, newFieldsOf`

Definition 13 (Object-Type Semantics)**.** $o \in [\![\mathsf{object}(B)]\!]$ *iff* $o \in \mathsf{ObjVal}$ *and there exists an object branch* $b \in B$ *with* $o \in [\![b]\!]$*, where* $o \in [\![b]\!]$ *iff all of:*

- ***Required fields:*** *for every* $f \in \mathsf{flds}(b) \cup \mathsf{symflds}(b)$ *with* $\mathsf{opt}(f) = \mathsf{false}$*:* $\mathsf{has}(o, \mathsf{name}(f))$ *and, unless* $\mathsf{access}(f) = \mathsf{Phantom}$*,* $\mathsf{get}(o, \mathsf{name}(f)) \in [\![\mathsf{get}(f)]\!]$*.*
- ***Optional fields:*** *for every such* f *with* $\mathsf{opt}(f) = \mathsf{true}$*: if* $\mathsf{has}(o, \mathsf{name}(f))$ *then* $\mathsf{get}(o, \mathsf{name}(f)) \in [\![\mathsf{get}(f)]\!]$*.*

- ***Indexes:*** *an index constrains the props it* covers*: the string index covers every string key (of either* strKey *or* numKey *kind), the number index covers* numKey*-kind keys only, and the symbol index covers symbol keys — matching TS, where a string index signature also constrains numeric properties, and the implementation's numeric lookup (`numberIndex ?? stringIndex`). For each index* $\mathsf{idx}_\kappa(b) = \mathsf{Some}(f)$ *and every* $p \in \mathsf{props}(o)$ *covered by it and not named by a field of* b*:* $\mathsf{get}(o, p) \in [\![\mathsf{get}(f)]\!]$.
- ***Call/new:*** *if* $\mathsf{call}(b) = \mathsf{CAbsent}$ *then* $\neg\mathsf{callable}(o)$*; if* $\mathsf{call}(b) \in \{\mathsf{CPresent}(\sigma), \mathsf{CFunction}\}$ *then* $\mathsf{callable}(o)$*;* CUnknown *imposes no constraint. Signatures themselves are opaque (their behavioral meaning is bundled into assumption* ***L-SUB****). Same for* new *with* constructible.
- ***Brand:*** *if* $\mathsf{isArr}(b) = \mathsf{Some}(x)$ *then* $\mathsf{isArrayV}(o) = x$.

The semantics is extensible*: properties not mentioned are unconstrained. A* Phantom *field constrains only* presence*, not the value — it records knowledge obtained from a runtime* `in` *check when no declaration is known; the implementation additionally makes phantoms error at access and be invisible to subtyping, which only* restricts *programs and so needs no semantic counterpart. The same restrict-only argument covers* Never *(produced when two same-named private fields with different owners meet: accessing it is a static error, but as a membership constraint it is an ordinary field). Note that membership constrains values via* get*-types only; the* set *component governs which writes the checker accepts and participates in soundness through subtyping and mutation, not through membership (assumption* ***L-MUT****). That governance creates a second obligation, cross-view coherence: a checked field or index write through one mutable view preserves the membership guarantees of every live alias of the same runtime object (****L-MUT-ALIAS****; its call-mediated counterpart is the row L-MUT-ALIAS-CALL) — which holds only while every mutable view's* set*-types stay at or below the object's actual write capability. For recursive declarations the membership conditions recurse — a field's* $[\![\mathsf{get}(f)]\!]$ *can mention* $[\![b]\!]$ *again — and every recursive occurrence is positive, so least and greatest solutions exist (Knaster–Tarski);* $[\![\cdot]\!]$ *is the* greatest *solution, the reading under which a cyclic value (`const x = {}; x.next = x`) inhabits its recursive type. This choice is registry row* ***SEM-GFP****.*

impl: `TType.sk / TObject (interpretation comment), TFields`

Definition 14 (Write Authorization)**.** *A view* b *(an object branch, as recorded for a local or a field)* authorizes *the write of value* v *at key* p *iff* $v \in [\![\mathsf{setty}(b, p)]\!]$*, where* $\mathsf{setty}(b, p)$ *is the resolved* set*-type of* p *in* b*: the intersection* $\bigsqcap f.\mathsf{set}$ *of a named field* f*, else the covering index's* set*-type resolved as at lvalue sites (the number index falling back to the string index);* $\mathsf{setty}(b, p) = \bot$ *when* p *has no writable field or covering index, and for a readonly field (*$\mathsf{set} = \bot$*) — either way* b *authorizes nothing at* p*. Optionality does not weaken authorization: an accepted write makes* p *present, and the stored value must satisfy the field's get-clause of Definition 13. The checker realizes this definition at assignment sites — the right-hand side is checked against* setty *by subtyping (****L-SUB****) — and writes mediated by method calls are covered by the signature semantics (E-SIG), not by* setty*. This is the corpus's definition of a view's write permissions, registry row* ***SET-SEM****;* ***L-MUT-ALIAS*** *is the coherence property over it.*

impl: `TType.sk / TFields::resolveFieldType (isLValue); TypeCheckExpr.sk / assign`

3.3 The union spine

Definition 15 (STypes)**.** STypes *is a finite, tag-indexed set: at most one* SType *per tag;* $[\![S]\!] = \bigcup_{s \in S} [\![s]\!]$*. Write* $S[k]$ *for the representative at tag* k *if present.*

impl: `TType.sk / STypes (InternalKeyedArrayMap keyed by STypeTag)`

Precision within a tag is carried by payloads: literal sets for primitives and object-branch sets for objects. A union of object types is represented exactly — one object branch per alternative — so merging alternatives loses no precision at the object tag.

Definition 16 (Instances)**.** TInst *is the abstract instance layer: type variables α (`TVar`, carrying a set of constraint types), unexpanded named instantiations (`TSInst`), unexpanded intersections (`TInter`), and delayed array-literal forms (`TArrayLit`).* TInsts *is a finite set of instances with $[\![I]\!]_\rho = \bigcup_{i\in I}[\![i]\!]_\rho$. Instance semantics is* parametrized by a substitution state ρ *(Section 6). The denotation of a type-variable occurrence α^{cs} (identifier α carrying the occurrence-level constraint set cs) is*

$$[\![\alpha^{cs}]\!]_\rho = \mathcal{E}_\rho(\alpha) \cap \bigcap_{c\in cs} [\![c]\!]_\rho, \qquad \mathcal{E}_\rho(\alpha) = \begin{cases} \emptyset & \alpha \notin \mathrm{dom}(\rho), \\ [\![t]\!]_\rho & \rho(\alpha) \in \{\mathsf{SVOpen}(t), \mathsf{SVClosed}(t)\}. \end{cases}$$

A binding type t may itself mention type variables, so this equation joins the recursive membership conditions of Definition 13; every recursive occurrence is positive, and the solution is the greatest one (registry row SEM-GFP). An unbound `TVar` thus denotes an empty inference lower bound, and `TVar` is the model's only inference *variable kind. Rigid abstraction is carried by* TParam*, not by a variable kind: a* skolem *is a synthetic* TParam *(named for its origin, e.g. `Box#T`, carrying the declared upper bound as its constraint) introduced where a type argument is fixed by a runtime object but hidden from the program — rule* R-Inst*'s hidden instance arguments; skolems live in clause var-sets like any parameter and never enter the substitution. Their interpretation is a definition of the model, registry row **SKOLEM-SEM**: a view installed by* R-Inst *is read under a* per-view witness assignment *mapping each of its skolems to the scrutinee object's actual hidden argument, chosen once when the view is installed. The assignment is a constant of the view's denotation — not open inference state — and it is stable because nothing can bind a skolem afterwards: skolems are not inference variables, the substitution never binds them, and skolem generation in clones is rejected. Rule* R-Inst*'s soundness proof (deep dive A) performs its existential step under exactly this assignment. Constraints belong to occurrences because `TVar::setConstraint` is functional; `Subst::get` realizes the displayed intersection when the occurrence is read. This equation is the corpus's definition of `TVar` denotation, registry row **TVAR-SEM**. Named instances and intersections satisfy the expansion axiom **L-EXP**.*

impl: `TType.sk / TInst, TVar, TSInst, TArrayLit; TTypeInter.sk / TInter; TTypeSubst.sk / genSkolemTParam`

Definition 17 (Clause, Clauses, TType)**.**

A clause $\langle V \mid S\rangle$ pairs type parameters $V \subseteq_{\mathrm{fin}}$ TParam with $S \in \mathsf{STypes}\cup\{\top\}$; clauses cannot be bottom (an empty STypes *is dropped at construction).* Clauses *is a map from var-sets to clauses, at most one per key, with the invariant that if the top clause $\langle\emptyset \mid \top\rangle$ is present it is alone.* $\mathsf{TType} = (I, C)$ *with $I \in$ TInsts, $C \in$ Clauses.*

$$[\![\langle V \mid S\rangle]\!]_\rho = \bigcap_{v\in V} [\![v]\!]_\rho \cap [\![S]\!]_\rho \qquad [\![C]\!]_\rho = \bigcup_{c\in C} [\![c]\!]_\rho \qquad [\![(I,C)]\!]_\rho = [\![I]\!]_\rho \cup [\![C]\!]_\rho$$

where $[\![\top]\!]_\rho = \mathsf{Val}$.

A clause may have $V \neq \emptyset$ with $S = \top$ (a type bounded only by its parameters); only $\langle\emptyset \mid \top\rangle$ is the top clause.

impl: `TType.sk / Clause, Clauses, TType`

3.4 Operation interfaces

Refinement and constant removal are built from a small set of operations whose soundness this document assumes as assumptions and uses freely:

Axiom 18 (L-UNION, L-INTER, L-INTER-BOT)**.** *For all t_1, t_2 and states ρ: $[\![t_1]\!]_\rho \cup [\![t_2]\!]_\rho \subseteq [\![\mathsf{union}(t_1, t_2)]\!]_\rho$. Intersection has two separately tracked cases. If $\mathsf{inter}(t_1, t_2) \neq \bot$, then, conditional on bottom-faithfulness of the recursive intersections used to construct it, $[\![t_1]\!]_\rho \cap [\![t_2]\!]_\rho \subseteq [\![\mathsf{inter}(t_1, t_2)]\!]_\rho$ (**L-INTER**). If $\mathsf{inter}(t_1, t_2) = \bot$, then $[\![t_1]\!]_\rho \cap [\![t_2]\!]_\rho = \emptyset$ (**L-INTER-BOT**), which is both the recursive premise of L-INTER and the premise refinement needs to drop a branch. Thus unconditional containment requires both rows; the second is* VIOLATED *in the registry. The same case split applies at the* STypes *and* object *levels.*

impl: `TTypeUnion.sk; TTypeInter.sk; TType.sk / STypes::union, TObject::inter`

Axiom 19 (L-EXP: Expansion)**.** *Expansion rewrites a type to an instance-plus-clauses view using the current substitution state, possibly evolving that state ($\rho \to \rho'$, e.g. sealing a fresh covariantly-read variable to top). We assume $[\![t]\!]_\rho \subseteq [\![\mathsf{expand}(t)]\!]_{\rho'}$ and that residual instances (those left in the instance layer after expansion) keep their semantics. All refinement operators run on the expanded view. Expansion takes a mode (`expandTVars`); the containment is claimed for modes `None/Covariant` only — contravariant expansion is an under-approximating read belonging to **L-SUB** — and every refinement client uses those modes (deep dive C).*

impl: `TType.sk / TType::expand, expandToVectors, mapSTypes`

Axiom 20 (L-MONO: Refinement-Time State Monotonicity)**.** *The substitution evolutions performed* while refining a condition *(expansion reads, which may seal fresh variables to top, and exploratory intersections) preserve membership for the types* installed in the environment *at the evolution, and for the type the rule just produced: if $\rho \to \rho'$ is such an evolution, then for every $u \in \mathrm{range}(\Gamma)$, $v \in [\![u]\!]_\rho \implies v \in [\![u]\!]_{\rho'}$. Without this, narrowing one variable could silently invalidate the recorded types of* other *variables. Two remarks. First, deep dive E proves this axiom outright (given semantic monotonicity in variable envelopes and **L-UNION**); in particular there is no constraint-mutation threat from shared state — `TVar::setConstraint` is a* functional *update on a frozen value, so constraints are occurrence-level (registry row SETCONSTRAINT-FUN) — and the* range(Γ) *restriction, stated here for safety, is droppable. Second, the classic contravariance counterexample (function parameter positions) is deflected only because signatures are opaque to $[\![\cdot]\!]$ — membership never inspects parameter types (Definition 13). This is the refinement-time counterpart of Obligation 47 (which concerns inference-time growth), and it is used by every branch-soundness argument below.*

impl: `TTypeSubst.sk / Subst::get (seal-to-top); TypeCheckRefine.sk (cloned-subst intersections)`

4 Refinement (Control-Flow Narrowing)

4.1 The single refinement primitive

Nearly all narrowing in SKJS is performed by one primitive applied to a local variable's type; the exceptions (equality's whole-variable intersection and the connectives' scope merge) are called out after the framework lemmas and receive direct proofs. The primitive is a filter/map: a partial mapper transforms each SType member of each clause of the expanded type, and an optional default replaces each *top* clause.

Definition 21 (Memberwise Refinement)**.** *Let* $g : \mathsf{SType} \rightharpoonup \mathsf{SType}$ *be a partial, tag-preserving mapper and* $d \in \{\mathsf{None}\} \cup \{\mathsf{Some}(D) \mid D \in \mathsf{STypes}\}$ *an optional default. Define* $\mathsf{refine}^{g,d}(t)$ *as follows: expand* t *(axiom* ***L-EXP****) to residual instances* I *and clauses* C*; replace each clause* $\langle V \mid S\rangle$ *by*

$$\begin{cases} \langle V \mid D\rangle & \text{if } S = \top \text{ and } d = \mathsf{Some}(D) \\ \langle V \mid \top\rangle & \text{if } S = \top \text{ and } d = \mathsf{None} \\ \langle V \mid \{g(s) \mid s \in S,\ g(s) \text{ defined}\}\rangle & \text{otherwise,} \end{cases}$$

dropping clauses whose STypes *becomes empty; the result is* (I, C') *rebuilt with the clause union.*
impl: `TypeCheckRefine.sk / setLocalFilterMap; TType.sk / TType::mapSTypes`

Definition 22 (Predicate Soundness)**.** *Let* $P \in \mathcal{P}(\mathsf{Val})$*. A partial mapper* g *is* sound for P *if for every* $s \in \mathsf{SType}$ *and* $v \in [\![s]\!] \cap P$*:* $g(s)$ *is defined and* $v \in [\![g(s)]\!]$*. A default* $d = \mathsf{Some}(D)$ *is* adequate for P *if* $P \subseteq [\![D]\!]$*;* $d = \mathsf{None}$ *is always adequate. Object-typed members make these denotations state-indexed (their field types may mention instances); throughout,* $[\![s]\!], [\![g(s)]\!], [\![D]\!]$ *are read at the post-refinement state* ρ'*. A transformer* $R : \mathsf{TType} \to \mathsf{TType}$ *is* predicate-sound for P *if* $[\![t]\!]_\rho \cap P \subseteq [\![R(t)]\!]_{\rho'}$ *for all* t *(with* $\rho \to \rho'$ *the state evolution of* R*).*

Lemma 23 (Memberwise Refinement is Predicate-Sound)**.** *If* g *is sound for* P *and* d *adequate for* P*, then* $\mathsf{refine}^{g,d}$ *is predicate-sound for* P*.*

Proof. Let $v \in [\![t]\!]_\rho \cap P$. By **L-EXP**, $v \in [\![(I, C)]\!]_{\rho'}$ for the expanded view. If $v \in [\![I]\!]$, the instance layer is unchanged and v survives. Otherwise $v \in [\![\langle V \mid S\rangle]\!]$ for some clause. If $S = \top$: with $d = \mathsf{None}$ the clause is unchanged; with $d = \mathsf{Some}(D)$, $v \in \bigcap_V [\![\cdot]\!]$ still, and $v \in P \subseteq [\![D]\!]$, so $v \in [\![\langle V \mid D\rangle]\!]$. If $S \neq \top$: $v \in [\![s]\!]$ for some $s \in S$, and by soundness of g, $g(s)$ is defined with $v \in [\![g(s)]\!]$, so the clause (which is then nonempty) retains v; clauses dropped for emptiness never contained v. Tag preservation of g is what makes the image a well-formed STypes (at most one member per tag), so no two survivors merge or clobber. Rebuilding with clause union only enlarges (**L-UNION**). □

Lemma 24 (Identity is Sound)**.** *The identity transformer is predicate-sound for every* P*.* □

Beyond the primitive. Two rule families do not factor through $\mathsf{refine}^{g,d}$: equality narrowing on a whole variable, which *replaces* the type by an intersection (Rule R-Eq); and the connective rules, which merge two refined scopes (Rule R-And/R-Or). They get direct proofs.

4.2 Environments, evaluation, and branch soundness

Definition 25 (Environments and Stores)**.** Γ *maps the* local *variables of the current scope to* TType*;* σ *maps locals to* Val*. Write* $\sigma \models_\rho \Gamma$ *when* $\sigma(x) \in [\![\Gamma(x)]\!]_\rho$ *for all* $x \in \mathrm{dom}(\Gamma)$*. Only locals are narrowed: the implementation's refinement primitive is a no-op for bindings not in the local scope (module-level declarations), since narrowing a global is unsound under concurrent mutation through closures or aliases. (Assumption* ***L-LOCAL****.)*
impl: `TypeCheckRefine.sk / setLocalFilterMap (non-local early return)`

Definition 26 (Evaluation of Condition Forms)**.** *For each condition form* e *of the master list we define relations* $\sigma \models_{\mathsf{true}} e$ *and* $\sigma \models_{\mathsf{false}} e$*, capturing the runtime outcomes under which the corresponding branch is taken. Evaluation of subexpressions is assumed not to modify* σ *(assumption* ***L-PURE****); forms whose evaluation would throw (e.g.* `"p" in 3`*) satisfy neither relation, which makes any narrowing of an unreachable branch vacuously sound.*

- $\sigma \models_b (e_1 \texttt{ === } e_2)$ *iff* $v_1 \approx v_2$ *(Definition 4) exactly when* $b = \mathsf{true}$*, where* v_1, v_2 *are the values of* e_1, e_2*.* `!==` *swaps* b*.*

- $\sigma \models_b (\texttt{typeof } x \texttt{ === } k)$ *iff* $\sigma(x) \in P_{\mathsf{typeof}=k}$ *exactly when* $b = \mathsf{true}$*, with*

$$P_{\mathsf{typeof}=\texttt{"undefined"}} = \{v \mid \mathsf{rtag}(v) \in \{\mathsf{undef}, \mathsf{void}\}\}$$
$$P_{\mathsf{typeof}=\texttt{"object"}} = [\![\texttt{null}]\!] \cup \{o \in \mathsf{ObjVal} \mid \neg\mathsf{callable}(o)\}$$
$$P_{\mathsf{typeof}=\texttt{"function"}} = \{o \in \mathsf{ObjVal} \mid \mathsf{callable}(o)\}$$

  and $P_{\mathsf{typeof}=k}$ *the evident tag set for* $k \in \{\texttt{"boolean"}, \texttt{"number"}, \texttt{"string"}, \texttt{"bigint"}, \texttt{"symbol"}\}$*.*

- $\sigma \models_{\mathsf{true}} (\texttt{"p" in } x)$ *iff* $\sigma(x) \in \mathsf{ObjVal}$ *and* $\mathsf{has}(\sigma(x), p)$*;* $\models_{\mathsf{false}}$ *iff* $\sigma(x) \in \mathsf{ObjVal}$ *and* $\neg\mathsf{has}(\sigma(x), p)$*. (A non-object* $\sigma(x)$ *throws.)*

- $\sigma \models_{\mathsf{true}} (x \texttt{ instanceof } C)$ *iff* $\sigma(x) \in P_{\mathsf{inst}}(C) \subseteq \mathsf{ObjVal}$ *(assumption* ***L-INST****).*

- $\sigma \models_{\mathsf{true}} \texttt{Array.isArray}(x)$ *iff* $\sigma(x) \in \mathsf{ObjVal}$ *and* $\mathsf{isArrayV}(\sigma(x))$*;* $\models_{\mathsf{false}}$ *is the complement over all of* Val*.*

- $\sigma \models_b (\texttt{!}e)$ *iff* $\sigma \models_{\neg b} e$*.*

- $\sigma \models_{\mathsf{true}} (e_1 \texttt{ \&\& } e_2)$ *iff* $\sigma \models_{\mathsf{true}} e_1$ *and* $\sigma \models_{\mathsf{true}} e_2$*;* $\sigma \models_{\mathsf{false}} (e_1 \texttt{ \&\& } e_2)$ *iff* $\sigma \models_{\mathsf{false}} e_1$*, or* $\sigma \models_{\mathsf{true}} e_1$ *and* $\sigma \models_{\mathsf{false}} e_2$*. Dually for* `||`*.*

- $\sigma \models_b x$ *(truthiness of a variable) iff* $\mathsf{truthy}(\sigma(x)) = b$*; likewise* $\sigma \models_b x.f$ *iff* $\sigma(x) \in \mathsf{ObjVal}$*,* $\mathsf{has}(\sigma(x), f)$ *or the read yields* `undefined`*, and the read value's truthiness is* b*.*

Definition 27 (Branch Soundness)**.** *A rule deriving* $\Gamma \vdash e \overset{b}{\rightsquigarrow} \Gamma'$ *is* branch-sound *if* $\sigma \models_\rho \Gamma$ *and* $\sigma \models_b e$ *imply* $\sigma \models_{\rho'} \Gamma'$*, where* $\rho \to \rho'$ *is the substitution evolution performed by the rule's refinement operators (an evolution of the kind covered by* ***L-MONO****).*

Lemma 28 (Predicate-Sound Update is Branch-Sound)**.** *If* R *is predicate-sound for* P*, the update* $\Gamma' = \Gamma[x \mapsto R(\Gamma(x))]$ *is branch-sound for any* e, b *with* $\sigma \models_b e \implies \sigma(x) \in P$*.*

Proof. For unchanged variables y: $\sigma(y) \in [\![\Gamma(y)]\!]_\rho$, and **L-MONO** carries this to ρ' — this step is *not* free, as the semantics of $\Gamma(y)$ is state-indexed. For x: $\sigma(x) \in [\![\Gamma(x)]\!]_\rho \cap P \subseteq [\![R(\Gamma(x))]\!]_{\rho'}$ by predicate-soundness of R. □

4.3 Master list

Each row corresponds to *one dispatch arm* of `evalCond` (`TypeCheckRefine.sk`); the maintenance check of Appendix B keeps this correspondence mechanical. "id" means that control-flow branch performs no narrowing.

Rule	Condition form	Branches	Impl (dispatch arm / handler)
R-Eq	x `===` e, x `===` y	both	`evalEquality`/`evalVarEquality`
R-EqUndef	x `===` `undefined`	both	`evalVarEquality` (undefined case)
R-EqField	$x.f$ `===` e (and $x.f$ `===` $y.g$)	both	`evalFieldEqualityType`, `narrowField`
R-Typeof	`typeof` x `===` k	both	`evalTypeofEquality`
R-IsArray	`Array.isArray(`x`)`	both	`evalIsArray`
R-Inst	x `instanceof` C	true only	`evalInstanceof`
R-In	e_k `in` x	both	`evalCond` (`In` arm)
R-Not	`!`e	both	`evalCond` (`Unop` arm)
R-And	e_1 `&&` e_2	both	`evalCond` (`BOLogicalAnd` arm)
R-Or	e_1 `||` e_2	both	`evalCond` (`BOLogicalOr` arm)
R-Truthy	x in condition position	both	`evalTruthiness`
R-TruthyField	$x.f$ in condition position	both	`evalFieldTruthiness`
R-Other	anything else	both (id)	`evalCond` final wildcard

Entry points that reduce to this list: `if`/`while` conditions, the ternary c `?` a `:` b (which narrows inside a and b exactly like `if`), and `&&`/`||` value positions call `evalCond`. A `switch` clause's body is checked under the true branch of the desugared label equality e `===` v; a clause entered *by fallthrough* additionally unions in the falling clause's exit scope (the fallthrough entry is a separate way in, not narrowed by the label), and the `default` body is checked under the false branch of *every* case equality — so it is narrowed to what no case matched (`TypeCheckStmt.sk`; `TypeCheckExpr.sk` block-entry narrowing). `console.assert`(e) applies the true branch of e — matched by call shape, first argument, without verifying `console` is the global (`TypeCheckExpr.sk`). The premise this clause would need — a recognized global assertion function that does not return when its condition is false — is registry row **L-ASSERT**, VIOLATED: a failed `console.assert` logs and continues, and a shadowing binding named `console` triggers the same refinement (witnessed both ways; issue report `console-assert-narrowing.md`). `for`-loop conditions do *not* narrow (`checkBool` only). Strict inequality dispatches to the same handlers with the target negated. The `instanceof` arm is guarded: it narrows only when the RHS names a class or interface with a known definition *and* the RHS's static type declares no custom `[Symbol.hasInstance]` member — such a class overrides the runtime membership test, so narrowing is suppressed entirely (`rhsDeclaresHasInstance`; registry row A-HASINST).

4.4 Rules and soundness

Throughout, b is the branch (true in the taken/then branch, false in the untaken/else branch) and updates go through Lemma 28 unless noted.

4.4.1 R-Eq: strict equality on a variable

For x `===` e with e not the literal `undefined` (that case is R-EqUndef), let τ_e be the checked type of e.

$$\frac{}{\Gamma \vdash (x \texttt{ === } e) \overset{\mathsf{t}}{\rightsquigarrow} \Gamma[x \mapsto \mathsf{inter}(\Gamma(x), \tau_e)]}\ \text{R-EqT} \qquad \frac{\mathsf{singleConst}(\tau_e) = \mathsf{Some}(c)}{\Gamma \vdash (x \texttt{ === } e) \overset{\mathsf{f}}{\rightsquigarrow} \Gamma[x \mapsto \mathsf{rmc}(\Gamma(x), c)]}\ \text{R-EqF}$$

where rmc is constant removal (Section 5) and singleConst succeeds only when τ_e is a single unit or single-literal primitive, whose denotation is then *contained in* a single $\approx$-class — equal to it for literal primitives and the `null` unit, a proper subset for the `undefined`/`void` units, whose cross-tag partner Definition 8 deliberately excludes. If singleConst fails, the false branch is the identity. When both sides are variables (or a variable and a field path), the rule applies to each side independently.
impl: `TypeCheckRefine.sk / evalVarEquality`

Lemma 29 (R-Eq is Branch-Sound)**.** *Assume $[\![\tau_e]\!]$ is $\approx$-closed (assumption **L-SEQ**).*

Proof. True branch. $\sigma \models_{\mathsf{true}}$ gives $\sigma(x) \approx v_e$, the value of e. By type correctness of the checker for e (assumption **L-TC**), $v_e \in [\![\tau_e]\!]$, and by $\approx$-closure $\sigma(x) \in [\![\tau_e]\!]$. So $\sigma(x) \in [\![\Gamma(x)]\!] \cap [\![\tau_e]\!]$. If the implemented intersection is non-bottom, **L-INTER** puts $\sigma(x)$ in it; if it is bottom, the same inhabitant contradicts **L-INTER-BOT**. Hence the installed intersection contains $\sigma(x)$, conditional on both rows. As in Lemma 28, the unchanged variables transport to the post-refinement state by **L-MONO** (this rule bypasses the memberwise primitive, so the step must be taken explicitly).
False branch. $\sigma(x) \not\approx v_e$, and every member of $[\![c]\!]$ is $\approx v_e$ (Lemma 5 for a literal constant, since $v_e \in [\![\tau_e]\!] = [\![c]\!]$; unit reflexivity otherwise), so $\sigma(x) \notin [\![c]\!]$; then $\sigma(x) \in [\![\Gamma(x)]\!] \setminus [\![c]\!] \subseteq [\![\mathsf{rmc}(\Gamma(x), c)]\!]$ by Theorem 40(R2). □

The $\approx$-closure hypothesis. It holds by construction for literal and full primitive types (Definition 8 takes the same-tag closure, using transitivity via Lemma 5, and $\approx$ is payload equality on non-numeric tags), for the `null` unit (payload equality at its own tag — null comparisons flow through this generic rule), and trivially for object types (reference equality). The corner where it fails is the cross-tag undef/void arm of $\approx$, in either direction: a τ_e denoting the `undefined` unit alone is not closed against void-tagged values, and dually for a `void`-typed RHS meeting an undefined-tagged value. The syntactic comparison x `=== undefined` takes the dedicated rule R-EqUndef, which treats the two tags jointly; a comparison against an undefined- or void-*typed* expression that is not the literal (`x === y` with `y: undefined`) reaches the generic rule, and the corner is *reachable*: a `void`-typed operand of strict equality draws only a *warning* ("An expression of type 'void' cannot be used here"), after which refinement runs. There the closure hypothesis is genuinely false — the generic rule's intersection treats the `void` and `undefined` units as disjoint, so the true branch of `x === y` with `x` of type `void` and `y` of type `undefined` narrows `x` to `never` although the runtime takes that branch with both operands holding `undefined`. This is the witnessed violation of assumption **L-SEQ** (issue report `deep-dives/issues/void-undefined-equality-never.md`); see the worked example in Section 7.5.

State caveat. The implementation computes the intersection under a *cloned* substitution and installs the resulting type against the main one; assumption **L-CLONE** records the stability requirement this creates.

4.4.2 R-EqUndef: comparison with `undefined`

Mapper g_b: keep s iff $(\mathsf{tag}(s) \in \{\mathsf{undef}, \mathsf{void}\}) = b$; no default. This is $\mathsf{refine}^{g_b, \mathsf{None}}$ on both branches. Soundness: $\sigma \models_b$ holds iff $(\sigma(x) \approx (\mathsf{undef}, \star)) = b$, and by Definition 4 that is exactly $(\mathsf{rtag}(\sigma(x)) \in \{\mathsf{undef}, \mathsf{void}\}) = b$ — the rule treats the two tags jointly precisely because $\approx$ identifies them. Apply Lemmas 23 and 28. (`null` comparisons are *not* special-cased here: they flow through R-Eq with $c =$ `null`.)
impl: `TypeCheckRefine.sk / evalVarEquality (undefined case)`

4.4.3 R-EqField: strict equality on a field path

For $x.f$ `===` e with RHS type τ_e and $c = \mathsf{singleConst}(\tau_e)$ (possibly undefined), the mapper acts only on object branches. For each branch β of each object member of $\Gamma(x)$, let τ_β be β's *resolved read type* for the *declared named* field f: $\mathsf{get}(f)$ if f is required, $\mathsf{union}(\mathsf{get}(f), \texttt{undefined})$ if optional (`maybeGetType`/`resolveFieldType` — the implementation resolves before narrowing, which is necessary for the proof below). If β has no declared field f — including when f is reachable only through an index signature — the object branch is kept unchanged (a sound over-approximation; no diagnostic is emitted). Let $\omega = \mathsf{inter}(\tau_\beta, \tau_e)$:

$$g_{\mathsf{true}}(\beta) = \begin{cases} \text{drop} & \omega = \bot \\ \beta[f.\mathsf{get} \mapsto \omega] & \text{else} \end{cases} \qquad g_{\mathsf{false}}(\beta) = \begin{cases} \beta & \omega = \bot \text{ or } c \text{ undefined} \\ \text{drop} & \mathsf{rmc}(\tau_\beta, c) = \bot \\ \beta[f.\mathsf{get} \mapsto \mathsf{rmc}(\tau_\beta, c)] & \text{else} \end{cases}$$

with the bookkeeping that narrowing out `undefined` also clears the field's optionality.[1] Non-object members are kept unchanged on both branches. The narrowed read type is written back as the field's get; when the field stays optional, the next read re-adds `undefined`, which is conservative.
impl: `TypeCheckRefine.sk / evalFieldEqualityType, narrowField`

Lemma 30 (R-EqField is Branch-Sound)**.** *Assume $[\![\tau_e]\!]$ is $\approx$-closed (**L-SEQ**), as in* R-Eq.

Proof. Let $v = \sigma(x) \in [\![\Gamma(x)]\!]$ and let the read value be $w = \mathsf{get}(v, f)$ if present, else the value `undefined`. For an object branch β with declared field f, membership of v in β gives $w \in [\![\tau_\beta]\!]$ *by construction of the resolved read type*: if f is present in v, Definition 13 gives $w \in [\![\mathsf{get}(f)]\!] \subseteq [\![\tau_\beta]\!]$ (**L-UNION**); if f is optional and absent, $w =$ `undefined` $\in [\![\tau_\beta]\!]$ because resolution unioned `undefined` in. (This is why resolving before narrowing is necessary: with the raw get-type, the true branch of `x.f === undefined` on an absent optional field would compute $\omega = \bot$ and wrongly drop the object branch.) *True branch:* $w \approx v_e$ and $v_e \in [\![\tau_e]\!]$ (**L-TC**), so $w \in [\![\tau_e]\!]$ by closure, and $w \in [\![\omega]\!]$ by **L-INTER**, with the bottom case excluded by **L-INTER-BOT**; in particular $\omega \neq \bot$, so v's object branch is not dropped. If f is present in v, the strengthened field is satisfied; if absent (possible only when $v_e \approx$ `undefined`), optionality is kept and the strengthened field imposes no constraint. *False branch:* $w \not\approx v_e$. If c is defined then every member of $[\![c]\!]$ is $\approx v_e$ (as in R-Eq), so $w \in [\![\tau_\beta]\!] \setminus [\![c]\!] \subseteq [\![\mathsf{rmc}(\tau_\beta, c)]\!]$ by Theorem 40(R2); since w inhabits it, the removal is not $\bot$ and the object branch survives with v in it. When c is the `undefined` unit, optionality is cleared: $w \not\approx$ `undefined` entails f is present in v, so the now-required field is satisfied. If $\omega = \bot$ then by **L-INTER-BOT** the field value could never strictly equal v_e (using closure of $[\![\tau_e]\!]$), and the object branch is kept unchanged, which is sound. Branches without a declared f and non-object members are unchanged (identity, Lemma 24). □

4.4.4 R-Typeof

Eight recognized operands; any other string is a static error (TS2367), and then no narrowing occurs. Writing b for the control-flow branch:

- $k \in \{$`"boolean"`, `"number"`, `"string"`, `"bigint"`, `"symbol"`$\}$: keep s iff $(\mathsf{tag}(s) = \mathsf{tag}(k)) = b$; default on the true branch only: the full primitive of kind k.
- $k =$ `"undefined"`: keep s iff $(\mathsf{tag}(s) \in \{\mathsf{undef}, \mathsf{void}\}) = b$; no default.

[1] The optionality-clearing test matches only the `undefined` unit, not `void`; subtracting a `void`-typed constant keeps optionality — conservative, hence sound.

- $k =$ `"object"`: on $b =$ true, keep `null`; drop other primitives; narrow each object branch by fnNarrow(false, false) (below). On $b =$ false: `null` dropped, other primitives kept, each object branch narrowed by fnNarrow(true, false). Default (true branch): $\{\texttt{null}\} \sqcup \mathsf{object}(\{\emptyset\})$.
- $k =$ `"function"`: on $b =$ true: drop all non-objects; narrow each object branch by fnNarrow(true, true). On $b =$ false: keep non-objects; narrow each object branch by fnNarrow(false, false). Default (true branch only): the global `Function` object type, whose single branch carries call = CFunction — so a clause $\langle V \mid \top \rangle$ becomes $\langle V \mid \mathsf{Function} \rangle$: a type parameter narrows to T & `Function` and `unknown` to `Function`, matching tsc. (The false branch has no default: "T minus functions" has no representation.)

Both object-branch cases use one transformer. fnNarrow($want, proven$) acts on an object branch β by its call mode (Definition 12): with $want =$ true, drop β if $\mathsf{call}(\beta) = \mathsf{CAbsent}$, keep it otherwise — strengthening CUnknown to CFunction only when *proven* (the `"function"` true branch has a runtime proof of [[Call]]; the `"object"` false branch does not and leaves CUnknown alone, which is conservative). With $want =$ false, drop β if $\mathsf{call}(\beta) \in \{\mathsf{CPresent}, \mathsf{CFunction}\}$, keep it otherwise with both modes stamped CAbsent. The construction invariant new $\geq$ CFunction $\implies$ call $\geq$ CFunction is why reading call alone suffices: a constructible branch is never mistaken for definitely-not-a-function.

impl: `TypeCheckRefine.sk / evalTypeofEquality; TType.sk / TFields::narrowByTypeofFunction`

Lemma 31 (R-Typeof is Branch-Sound)**.**

Proof. Tag cases are immediate from A0 and Definition 26; defaults are adequate since $P_{\mathsf{typeof}=k} \subseteq [\![\text{full } k]\!]$. For `"object"`, the default's object branch (`TObject::createEmpty`) has no fields or indexes and call = new = CUnknown, which constrains nothing, so adequacy is immediate. For `"function"`, the default replaces a top clause $\langle V \mid \top \rangle$ keeping its parameters, and its branch carries call = CFunction: every $o \in P_{\mathsf{typeof}=\texttt{"function"}}$ is callable and satisfies it (the `Function` branch has no other constraint — its brand field folds into the mode and is not a property), so adequacy holds. For fnNarrow, let $v \in \mathsf{ObjVal} \cap [\![\beta]\!]$. If callable($v$) (the $want =$ true predicates): $\mathsf{call}(\beta) \neq \mathsf{CAbsent}$ by Definition 13, so β survives, and when the mode is strengthened to CFunction under *proven*, v satisfies it by its callability — the runtime proof is exactly the strengthening's premise; without *proven* the mode is unchanged and v satisfies it a fortiori. If $\neg$callable(v) (the $want =$ false predicates): $\mathsf{call}(\beta) \notin \{\mathsf{CPresent}, \mathsf{CFunction}\}$, so β survives, and the stamped CAbsent modes are satisfied — v is not callable, hence by the constructible $\implies$ callable fact not constructible either. The mode rewrites preserve all other constraints of β and are tag-preserving, so Lemma 23 applies. □

4.4.5 R-IsArray

True branch: drop non-object members; each object branch is intersected with the single object branch of object(ReadonlyArray$\langle\top\rangle$) and its brand set to Some(true); an object branch whose intersection is $\bot$, or that was already branded Some(false), is dropped. (The `ReadonlyArray` stub itself carries the brand as a declared field `readonly __SKJS_isArray: true`, so the brand-conflict drop already occurs inside the intersection.) False branch: object branches have the brand set to Some(false); a branch already branded Some(true) is dropped; non-objects are kept.

impl: `TypeCheckRefine.sk / evalIsArray; TTypeInter.sk / TFields::interIsArray`

Lemma 32 (R-IsArray is Branch-Sound)**.**

Proof. True branch: $\sigma \models_{\mathsf{true}}$ gives $v = \sigma(x) \in \mathsf{ObjVal}$ with $\mathsf{isArrayV}(v)$. By assumption **L-ISARR**, $v \in [\![\mathsf{object}(\mathsf{ReadonlyArray}\langle\top\rangle)]\!]$; by **L-INTER** (object level) v inhabits a non-bottom intersected branch, while **L-INTER-BOT** excludes bottom because v inhabits both operands; therefore v inhabits the implemented intersection, and setting the brand to $\mathsf{Some}(\mathsf{true})$ is satisfied by $\mathsf{isArrayV}(v)$. False branch: if $v \in \mathsf{ObjVal}$ then $\neg\mathsf{isArrayV}(v)$ satisfies brand $\mathsf{Some}(\mathsf{false})$ and contradicts membership in a $\mathsf{Some}(\mathsf{true})$-branded branch, so dropping those loses nothing; non-objects pass through. □

Mutability note. Intersecting with *Readonly*Array rather than `Array` is a deliberate precision/soundness choice documented in the implementation: a readonly-array-typed value may be a non-array object carrying the right properties in general, but under isArrayV the brand is known; what is *not* known is mutability, so the mutable interface is added only where the original branch already had it.

4.4.6 R-Inst: `instanceof`

Fires only on the true branch, and only when the RHS names a class or interface with a known definition (a Function-typed value as RHS narrows nothing) that declares no custom `[Symbol.hasInstance]` (the dispatch guard; registry row A-HASINST). Let T_C be the object-typed instance of C at *skolem* arguments — one synthetic rigid TParam per class parameter (Definition 16), carrying the declared bound: the runtime object fixes each argument, but the program cannot name it. Mapper: drop non-objects; each object branch β is first probed for *preservation* — if β is already a subtype of C's fields at fresh probe arguments (created in a temporary clone, per branch, so the main substitution never sees them), β is kept *unchanged*, retaining its own type arguments rather than intersecting them away — otherwise β is intersected with T_C's fields (dropping empty intersections), and the intersection retains β's readonly fields readonly: the class view's mutable fields must not re-grant a write the scrutinee's type withheld. Default: T_C itself.

impl: `TypeCheckRefine.sk / evalInstanceof, rhsDeclaresHasInstance, evalCond (InstanceOf arm); TType.sk / TFields::keepReadonlyFrom; TEnv.sk / makeSkolemTArgsOf`

Lemma 33 (R-Inst is Branch-Sound)**.** *Given assumption* ***L-INST*** *($P_{\mathsf{inst}}(C) \subseteq [\![T_C]\!]$, with T_C as instantiated at the narrowing site), the true-branch rule is branch-sound; the false branch is the identity.*

Proof. Let $v = \sigma(x)$ with $\sigma \models_{\mathsf{true}}$, so $v \in P_{\mathsf{inst}}(C) \subseteq \mathsf{ObjVal}$. Dropping non-object members is sound by A0 (values of a non-object member carry that member's tag, so v is not in it). A preserved branch is the identity (Lemma 24). For a rewritten branch, $v \in [\![T_C]\!]$ by **L-INST**. The non-bottom case is contained by **L-INTER** (object level), and the bottom case is excluded by **L-INTER-BOT**, since v inhabits both operands; hence v inhabits the implemented intersection, which is in particular not $\bot$; forcing a field's set-type back to $\bot$ only reduces the writes the view authorizes, and membership constrains values through get-types alone (Definition 13), so the readonly restoration preserves v's membership. The default $D = \{T_C\}$ is a singleton STypes (T_C is a `TObject` built by `newObject`) and is adequate by **L-INST** directly. The skolem arguments are rigid parameters — nothing about them lives in ρ, so no later growth threatens the installed type; the preservation probes run on clones and commit nothing (Obligation 48). Apply Lemmas 23 and 28. □

4.4.7 R-In

The key expression's *type* determines the refinement: if it denotes a single string *or numeric* literal p or a single (unique) symbol q, narrowing proceeds with key $\pi \in \{p, q\}$ — a numeric literal is

converted to its string spelling and the key kind is hardcoded to string, so the true branch's index lookup below consults the *string* index even for a numeric key — sound under the covering semantics of Definition 13, since a string index covers numeric-kind keys too; the imprecision is only that a number index is not consulted. If the key type is a multi-literal or non-literal string/number/symbol type, object members are kept unchanged — but primitive members are still dropped on the true branch; if it is not a string/number/symbol type at all, the checker raises a *fatal* TS2360 and no narrowing (or further checking of the definition) occurs. In every non-fatal case the true branch drops primitive members (evaluation of `in` on a primitive throws; Definition 26). With a single key π, per object branch β:

- **True branch:** if β has field π: clear its optionality. Else if β has a non-synthetic index *covering* π with field f_{idx}: when $f_{\text{idx}}.\mathsf{get} = \bot$ (a never-valued index, the form `Closed<T>` installs on uncovered keys), *drop* the object branch — the index proves π absent from every member, so this branch cannot take the true edge; otherwise add the required field $\pi \mapsto f_{\text{idx}}$. Else: add a required *phantom* field π ($\mathsf{get} = \top$, $\mathsf{set} = \bot$, $\mathsf{access} = \mathsf{Phantom}$).
- **False branch:** if β has required π: drop the object branch. If optional π: remove the field from β. Else: keep.

There is no default: a top-clause type stays top on both control-flow branches.
impl: `TypeCheckRefine.sk / evalCond (In arm), inLhsFieldsFromType`

Lemma 34 (R-In is Branch-Sound)**.**

Proof. True branch: $v = \sigma(x) \in \mathsf{ObjVal}$ with $\mathsf{has}(v, \pi)$. Primitives cannot contain v (A0), so dropping them is sound. If β had optional π, presence upgrades v's satisfaction to the required form; the value constraint is unchanged. If the field is added from an index covering π: membership of v in β already constrains $\mathsf{get}(v, \pi) \in [\![f_{\text{idx}}.\mathsf{get}]\!]$ (Definition 13, index clause — π is covered and was not named), so v satisfies the named copy; in the never-valued case that constraint reads $\mathsf{get}(v, \pi) \in [\![\bot]\!] = \emptyset$, which is unsatisfiable — v was never in β, so dropping the branch loses nothing. A phantom field constrains presence only, which holds. False branch: $v \in \mathsf{ObjVal}$ with $\neg\mathsf{has}(v, \pi)$ contradicts a required π — including a required *phantom*, which still demands presence — so v was never in dropped object branches. Removing an optional field π is *not* monotone in general (the removal re-exposes π to the object branch's index constraint, which the named field had exempted), but it is sound here: v lacks π, so the re-exposed index constraint on π is vacuous for v, and constraints on other keys are unchanged. Primitives kept is identity. □

Remark. The branch drop trusts the never-valued index as a truthful absence fact about v. Within this lemma that is nothing extra: the store satisfies the environment by hypothesis, and membership at a never-valued index is already unsatisfiable for a value carrying the key. What the lemma does *not* establish is that environment satisfaction survives admission into a closed view — that no extensible source is ever accepted into one. That is registry row **L-SUB-CLOSED-ABSENCE** (deep dive D's absence corollary): a prerequisite of the future preservation theorem and of deep dive H's cross-view argument, not a premise of this proof. R-In consumes the semantic absence fact; the row is what will make that fact survive the program.

4.4.8 R-Not, R-And, R-Or

$$\frac{\Gamma \vdash e \overset{\neg b}{\rightsquigarrow} \Gamma'}{\Gamma \vdash \texttt{!}e \overset{b}{\rightsquigarrow} \Gamma'}\ \text{R-Not} \qquad \frac{\Gamma \vdash e_1 \overset{\mathsf{t}}{\rightsquigarrow} \Gamma_1 \qquad \Gamma_1 \vdash e_2 \overset{\mathsf{t}}{\rightsquigarrow} \Gamma_2}{\Gamma \vdash (e_1 \,\texttt{\&\&}\, e_2) \overset{\mathsf{t}}{\rightsquigarrow} \Gamma_2}\ \text{R-AndT}$$

$$\frac{\Gamma \vdash e_1 \overset{\mathsf{f}}{\rightsquigarrow} \Gamma_1 \qquad \Gamma \vdash e_1 \overset{\mathsf{t}}{\rightsquigarrow} \Gamma' \qquad \Gamma' \vdash e_2 \overset{\mathsf{f}}{\rightsquigarrow} \Gamma_2}{\Gamma \vdash (e_1 \,\texttt{\&\&}\, e_2) \overset{\mathsf{f}}{\rightsquigarrow} \Gamma_1 \sqcup \Gamma_2}\ \text{R-AndF}$$

and dually R-OrF (sequential) and R-OrT (merge of "e_1 true" and "e_1 false then e_2 true"). Here $(\Gamma_1 \sqcup \Gamma_2)(x) = \mathsf{union}(\Gamma_1(x), \Gamma_2(x))$ is the pointwise scope merge. (The implementation merges only the bindings the sub-scopes updated: a variable narrowed in just one sub-scope is unioned with the *parent* scope's type — which equals the pointwise union, since the un-narrowed side's value is the parent's.)

impl: `TypeCheckRefine.sk / evalCond (Unop / BOLogicalAnd / BOLogicalOr arms); TypeCheckScope.sk / Scope::merge via TypeCheckEnv.sk / mergeScope`

Lemma 35 (Merge Soundness)**.** *If $\sigma \models \Gamma_1$ or $\sigma \models \Gamma_2$, then $\sigma \models \Gamma_1 \sqcup \Gamma_2$.*

Proof. Pointwise from **L-UNION**. The domains agree ($\mathrm{dom}(\Gamma_1) = \mathrm{dom}(\Gamma_2)$) because every rule is an update $\Gamma[x \mapsto \cdot]$ of an existing local — refinement never adds or removes bindings. The merged environment, like all post-refinement environments, is interpreted at the final state; the premises' satisfaction transports to it by **L-MONO**. □

Lemma 36 (Connectives are Branch-Sound)**.** *If all premise derivations are branch-sound, so are* R-Not, R-And, R-Or *— for* both *branches.*

Proof. R-Not is immediate from Definition 26. R-AndT composes directly: both conjuncts hold and the two refinements chain. For R-AndF: by Definition 26, $\sigma \models_{\mathsf{false}} e_1$ — giving $\sigma \models \Gamma_1$ — or $\sigma \models_{\mathsf{true}} e_1$ and $\sigma \models_{\mathsf{false}} e_2$ — giving $\sigma \models \Gamma'$ then $\sigma \models \Gamma_2$. Either way Lemma 35 concludes. The `||` cases are dual. Note the exactness of the short-circuit semantics in Definition 26 is what licenses refining *both* control-flow branches of both connectives. □

4.4.9 R-Truthy, R-TruthyField

Definition 37 (Truthiness Residue)**.** *For $b \in \{\mathsf{true}, \mathsf{false}\}$, the partial mapper $\mathsf{tres}_b : \mathsf{SType} \rightharpoonup \mathsf{SType}$ is:*

- $\mathsf{unit}(\texttt{void})$*: kept unchanged on both branches (*`void`* is treated as non-observable and never narrowed);*
- $\mathsf{unit}(\texttt{undefined}), \mathsf{unit}(\texttt{null})$*: dropped if b, kept if $\neg b$;*
- *objects and symbols: kept if b, dropped if $\neg b$;*
- *full* `boolean`*: the literal b;*
- *full* `string`*/*`number`*/*`bigint`*: kept unchanged on* both *branches;*
- *literal primitives $\mathsf{prim}(k, L)$: keep the literals $\ell \in L$ with $\mathsf{truthy}(\delta_k(\ell)) = b$; undefined if none remain.*

The value *residue* vres_b *differs in exactly two cases: full* `string` *and full* `bigint` *on* $b = \mathsf{false}$ *narrow to the literals* `""` *and* `0n` *(their unique falsy values); full* `number` *must stay wide because* NaN *has no literal type.*

impl: TType.sk / SType::truthinessResidue, truthinessValueResidue

Lemma 38 (Residues are Sound)**.** tres_b *is sound (Definition 22) for* $P_b = \{v \mid \mathsf{truthy}(v) = b\} \cup [\![\texttt{void}]\!]$*, and* vres_b *is sound for* P_b *as well.*

Proof. Case analysis over Definition 6. The two deliberate widenings (full string/number/bigint kept on both branches) only enlarge the result. The literal-filtering case is sound *only if* the implementation's lexeme-level falsiness test agrees with Definition 6 on every literal in L — this is the falsiness-exactness part (iii) of assumption **L-LIT**. String and within-precision numeric lexemes are canonical (escapes decoded, numerals normalized at parse time), so the test is exact there; the residual violation is the num-kind *beyond-precision* stratum (see the assumptions table): a zero-denoting underflow lexeme such as `1e-400` is classified truthy and would be dropped from the falsy branch. For $\mathsf{vres}_{\mathsf{false}}$: the only falsy string value is `""` and the only falsy bigint value is 0, so the narrowed literal contains every falsy member; number stays wide precisely because its falsy set $\{0, -0, \mathsf{NaN}\}$ is not expressible as literals. The $[\![\texttt{void}]\!]$ disjunct in P_b accounts for keeping `void` on both branches (for $b = \mathsf{false}$ it is redundant, since void values are falsy). □

R-Truthy is $\mathsf{refine}^{\mathsf{tres}_b,\mathsf{None}}$ on $\Gamma(x)$; branch soundness follows from Lemmas 23, 38 and 28 (note $\sigma \models_b x$ implies $\sigma(x) \in P_b$, and `void`-typed reads evaluate to `undefined`, which the kept `void` member covers). R-TruthyField applies the *type-level* residue ($\mathsf{refine}^{\mathsf{tres}_b,\mathsf{None}}$ on the field's type) through the same per-branch field update as R-EqField, dropping object branches whose residue is $\bot$ and clearing optionality on the true branch (the read value being truthy entails the field is present with a non-`undefined` value); its proof combines Lemma 30's structure with Lemma 38.

impl: `TypeCheckRefine.sk / evalTruthiness, evalFieldTruthiness; TType.sk / TType::truthinessPart`

Value positions of `&&`/`||`. The *type of the expression* e_1 `&&` e_2 uses $\mathsf{vres}_{\mathsf{false}}$ on e_1's type for the untaken-operand contribution (and dually for `||`); this is a typing rule rather than a refinement and inherits soundness from Lemma 38.

impl: `TType.sk / TType::truthinessValuePart`

5 Constant Removal

SKJS has no general type-difference operation: inference growth is structural (Section 6) and consumes no differences, and the only subtraction clients — destructuring defaults ($\tau \setminus \texttt{undefined}$) and `!==` narrowing — subtract a *single constant*. (The historical `TType::diff` and its per-level strata were deleted when growth became structural.)

Definition 39 (Constant Removal)**.** *A* constant c *is a unit or a single-literal primitive; its denotation* $[\![c]\!]$ *is* contained in *a single* $\approx$*-class — equal to it for literal primitives (for a numeric constant this may be the two-element class* $\{0, -0\}$*) and the* `null` *unit; a proper subset for the* `undefined/void` *units, whose cross-tag partner Definition 8 deliberately excludes.* $\mathsf{rmc}(t, c)$ *expands* t *and maps each clause's* STypes *elementwise:*

- $\mathsf{unit}(u)$ *with* $\mathsf{tag}(u) = \mathsf{tag}(c)$*: dropped;*
- $\mathsf{prim}(k, L)$ *with* $\mathsf{tag}(k) = \mathsf{tag}(c)$ *and* $L \neq \top$*: becomes* $\mathsf{prim}(k, L \setminus \{\ell_c\})$*, dropped if empty;*

- *full* `boolean` *minus a boolean literal: the complementary literal (exact);*
- *full* `number/string/bigint/symbol`*: kept unchanged (over-approximation);*
- *all other members (different tag, objects): kept unchanged.*

Top clauses are unchanged. The instance layer is unchanged.
impl: `TType.sk / TType::removeConstant; PrimLiterals.sk / diff`

Theorem 40 (Soundness of Constant Removal)**.** *For all t, c (and states, suppressed):*

(R1) $[\![\mathsf{rmc}(t,c)]\!] \subseteq [\![\mathsf{expand}(t)]\!]$*;*

(R2) $[\![t]\!] \setminus [\![c]\!] \subseteq [\![\mathsf{rmc}(t,c)]\!]$*.*

Moreover removal is exact *(*$[\![\mathsf{rmc}(t,c)]\!] = [\![\mathsf{expand}(t)]\!] \setminus [\![c]\!]$ *on the clause part) whenever the expansion has no top clause and every same-tag member is a unit, a literal primitive, or full* `boolean` *— assuming additionally the injectivity part (ii) of* ***L-LIT*** *(distinct normalized lexemes denote $\approx$-inequivalent values).*

Proof. (R1): every output member's semantics is a subset of the corresponding input member's ($L \setminus \{\ell_c\} \subseteq L$; the boolean complement is a subset of full boolean; kept members are equal). (R2): let $v \in [\![t]\!] \setminus [\![c]\!]$; by **L-EXP** $v \in [\![\mathsf{expand}(t)]\!]$. If v is in the instance layer or a top clause, it survives unchanged. Otherwise $v \in [\![s]\!]$ for a member s. If $\mathsf{tag}(s) \neq \mathsf{tag}(c)$, s is kept. If s is a unit of c's tag, then $s = c$ (the tag determines the unit) so $[\![s]\!] = [\![c]\!]$, contradicting $v \notin [\![c]\!]$ — vacuous. (A `void`-tagged v with c the `undefined` unit falls under the *different-tag* case: $\mathsf{unit}(\texttt{void})$ is kept, and $(\mathsf{void}, \star) \notin [\![c]\!]$ under the singleton unit denotations, so R2 is unaffected.) If s is a literal primitive, $v \approx (\mathsf{tag}(k), \delta_k(\ell))$ for some $\ell \in L$; if $\ell = \ell_c$ then $v \in [\![c]\!]$, contradiction — so $\ell \neq \ell_c$ and ℓ survives the lexeme diff, keeping v in the closure semantics. Note this step needs only that δ is a *function*, not injectivity: R2 is unconditional. If s is full boolean and c a boolean literal, v is the complementary literal. Full other primitives are kept whole. Exactness: a top clause is unchanged while $[\![\top]\!] \setminus [\![c]\!] \subsetneq [\![\top]\!]$, hence its exclusion; in the remaining listed cases each step removes exactly $[\![c]\!]$, using injectivity to ensure no *other* surviving lexeme denotes a member of $[\![c]\!]$. (R1 has no client among the soundness results in this document; its role is precision.) □

Subtracting from top. A top clause is left unchanged: `unknown` minus a constant is still `unknown`. Narrowing `unknown` by `!== null && !== undefined` therefore does not narrow at all — an incompleteness localized to the top-clause case of Definition 39 and to the residue rules (a top clause has no members to filter). Expressing a "non-nullish top" would still require a representation extension.

6 Inference: Substitution Growth and Validation

This section models the substitution state machine exactly and states the soundness architecture as explicit obligations.

6.1 The substitution state machine

Definition 41 (Substitution States)**.** *A substitution state ρ maps type-variable identifiers to* $\mathsf{SVOpen}(t)$ *(a growable lower bound) or* $\mathsf{SVClosed}(t)$ *(a sealed value), and carries a boolean* freeze *(the candidate-freezing flag). Exactly the* ordinary *identifiers (*`TVID`*) are* candidates *(*`TVarID: :`

`isCandidate`); field-inference (`TVIDForField`) and literal-commitment (`TVIDForLiteralCommitment`) identifiers are not *— the code classifies both as literal-form commitments, and they* are *sealed or widened by a covariant grow. Lemma 43 therefore protects ordinary inference candidates only.*
impl: `TTypeSubst.sk / SubstValue, Subst; TType.sk / TVarID::isCandidate`

Definition 42 (Transitions)**.** $\mathsf{grow}(\alpha, t, var)$ *and* $\mathsf{get}(\alpha, var)$ *act as follows (writing* $\vee$ *for `TType::union`):*

op	*state of* α	*condition*	*effect / result*
$\mathsf{grow}(t, \mathsf{contra})$	*unbound*		$\alpha \mapsto \mathsf{SVOpen}(t)$
	$\mathsf{SVOpen}(t_2)$	$\neg(\mathsf{cand}(\alpha) \wedge \mathsf{freeze})$	$\alpha \mapsto \mathsf{SVOpen}(t_2 \vee t)$
	$\mathsf{SVOpen}(t_2)$	$\mathsf{cand}(\alpha) \wedge \mathsf{freeze}$	*no-op*
	$\mathsf{SVClosed}(_)$		*no-op*
$\mathsf{grow}(t, \mathsf{cov})$	*unbound*		$\alpha \mapsto \mathsf{SVClosed}(t)$
	$\mathsf{SVOpen}(t_2)$	$\mathsf{cand}(\alpha)$	*no-op* (unconditional)
	$\mathsf{SVOpen}(t_2)$	$\neg\mathsf{cand}(\alpha)$	$\alpha \mapsto \mathsf{SVClosed}(t_2 \vee t)$
	$\mathsf{SVClosed}(_)$		*no-op*
$\mathsf{get}(\mathsf{cov})$	*unbound*		*seal* $\alpha \mapsto \mathsf{SVClosed}(\top)$*; return* $\top$
	$\mathsf{SVOpen}(t)$	$\mathsf{cand}(\alpha) \wedge \mathsf{freeze}$	*return* t *(no seal)*
	$\mathsf{SVOpen}(t)$	*otherwise*	*seal* $\alpha \mapsto \mathsf{SVClosed}(t)$*; return* t
	$\mathsf{SVClosed}(t)$		*return* t
$\mathsf{get}(\mathsf{contra})$	*unbound*		*return* $\bot$ *(no write)*
	$\mathsf{SVOpen}(t)$ *or* $\mathsf{SVClosed}(t)$		*return* t

Both gets *then intersect the result with* α*'s constraint types.*
impl: `TTypeSubst.sk / Subst::grow, Subst::get`

Lemma 43 (Determined Candidates Are Never Widened)**.** *For a candidate* α *bound to* $\mathsf{SVOpen}(v)$ *or* $\mathsf{SVClosed}(v)$*: (i) a covariant* grow *never changes* v*; (ii) under* freeze*, no* grow *or* get *changes* v *or its open/closed status; (iii) once* SVClosed*, the binding never changes again.*

Proof. By inspection of Definition 42: (i) the SVOpen/candidate/covariant row is an unconditional no-op and the SVClosed rows are no-ops; (ii) the two freeze-guarded rows are the only remaining writes touching a bound candidate, and both are suppressed; (iii) no transition rewrites an SVClosed binding. □

This is the write-site invariant behind the "covariant checks never widen a determined inference candidate" work (#713, #740): a covariant grow pushes an *expected* type — an upper bound — and the value of a candidate must come from its defining expression, with the comparison against the expected type happening in validation, not by overwriting the candidate.

Write-site audit. The lemma's premise — that Definition 42 captures *every* write to the binding map — has been checked against the source: the only assignments to `Subst.values` are in `grow`, `get`, and the bulk `restore`. External callers of `grow` (array-literal commitment in

`TArrayLitTyping.sk`, variadic and utility sites in `TVariadicTyping.sk` / `TypeCheckUtils.sk`) go through the same transitions; `TVar::setConstraint` (used by intersection) is a *functional* update returning a new `TVar` with an extended constraint set — constraints are occurrence-level, not shared state (registry row SETCONSTRAINT-FUN) — and is accounted for by the constraint intersection in get.

6.2 Growth as pre-seeding; validation decides

Definition 44 (Subtyping Check Architecture)**.** *`isSub`(t_1, t_2) proceeds as: (0) acceptance shortcuts that return before any growth — $t_1 = t_2$, and the reflexive `TSInst` arm (equal name, arguments equal after `peekRaw`); neither operation provides inference guidance. The `TSInst` arm's `peekRaw` reads an open candidate* without sealing it*, the unsealed-read instance of Obligation 47 (deep dive D); (1)* growTVars$(t_1, t_2, \mathsf{contra})$ *— grow variables of the to-be-grown side t_2 (the implementation's "open" side) from bound material in t_1; (2)* growTVars(t_2, t_1, cov) *— seal variables of t_1; (3) expand both sides under the (now grown) state and validate with `Clauses::isSub`; on failure,* restore *the substitution to its state before (1).* growTVars *itself: collects growable variables of the to-be-grown side (bare, inside intersections if unbound, and behind one-level alias expansion); checks each variable's declared constraints against the bound side* under freeze *(a query, per Lemma 43(ii)), rolling back on failure; grows only the disjuncts of the bound side not already covered by a non-variable branch of that side (each coverage probe is save/restored); finally descends structurally into matching object fields and call/new* return *types.*

impl: `TTypeSubtyping.sk / isSub_, growTVars, collectTVars`

Theorem 45 (Guidance vs. Validation)**.** *Assume the validation obligation **L-SUB**: if `Clauses::isSub` accepts under state ρ, then $[\![t_1]\!]_\rho \subseteq [\![t_2]\!]_\rho$ for the expanded views. Then no imprecision in* growTVars *— growing too little, too much, or in the wrong order — can cause `isSub` to accept an inclusion that is invalid* under the state ρ' that growth produced*. Growth affects only* completeness *(which valid programs are accepted) and the* solutions *chosen for inference variables. (That already-validated judgements are not retroactively weakened by later evolution is a separate matter: Lemma 43 for candidates, Obligations 47 and 48 in general; the failure-path parenthetical in the proof leans on Obligation 48.)*

Proof. Acceptance occurs only in step (3), whose verdict is by **L-SUB** a semantic inclusion *under the state that growth produced.* Growth selects the state; it never bypasses validation. (Failure paths restore the prior state, so failed exploration does not corrupt later checks — see Obligation 48.) □

The force of Theorem 45 is architectural rather than representational. What it makes explicit is that the meaning of *other, already-checked* judgements must not be retroactively weakened by later growth. That is exactly what Lemma 43 secures for candidates, and what the remaining obligations must secure in general:

Obligation 46 (Validation Soundness, **L-SUB**)**.** *`Clauses::isSub` (with its* SType*s, object-branch, field, index, and function strata, including getter/setter variance and the `seenSub`/`tenv.seenSub` cycle caches keyed by substitution version) is sound in the sense of Definition 44. This is the largest unproved component of the document.*

impl: `TTypeSubtyping.sk / Clauses::isSub, STypes::isSub, TObject::isSub, TFields::isSub, TFun::isSub`

Obligation 47 (Monotone Reads Under Growth (registry row OBL-MONOTONE))**.** *If a judgement was validated under ρ and inference later evolves $\rho \to \rho'$ by the transitions of Definition 42, the validated inclusion still holds under ρ'. For* SVClosed *bindings this is Lemma 43(iii); for*

SVOpen *bindings of* non-candidates*, unions grow the denotation and the obligation is that every consumer read them contravariantly (as lower bounds) until sealed.* This obligation is VIOLATED in two witnessed instances — *`subst.seenSub` memos recorded on unsealed candidate bounds inside `freezeCandidates` windows (order-dependent acceptance), and the reflexive `TSInst` arm of Definition 44 step (0), whose `peekRaw` is an exact-value read of an unsealed open candidate (not memo-mediated). The general statement is proved by deep dive E modulo exactly these residues; witnesses and fix directions: `deep-dives/issues/freeze-window-stale-memo-caches.md`, `deep-dives/issues/reflexive-tsinst-fastpath-unsealed-read.md`.*

Obligation 48 (No Commitment on Failure (registry row OBL-RESTORE))**.** *Every failing or exploratory path restores the substitution: `isSub` restores on validation failure; coverage probes and constraint-check failures inside* growTVars *restore; refinement's exploratory intersections run on clones (**L-CLONE**). The stale-cache regression class (#446) is guarded by keying cycle caches with the substitution version.*

7 Worked Examples

7.1 Literal narrowing

A worked case of R-Eq's false branch:

```
function f(x: 1 | 2): 2 {
  if (x === 1) { return 2; }
  return x;  // narrowed to 2
}
```

is accepted: `1 | 2` is prim(num, $\{1, 2\}$); the false branch of R-Eq applies rmc with $c = 1$, and Theorem 40 gives exactly prim(num, $\{2\}$).

7.2 Persisting incompleteness: subtracting from top

`unknown` is the top clause; R-Eq's false branch and the truthiness residues leave top clauses unchanged (Section 5), so `x !== null && x !== undefined` does not narrow `x: unknown`. This remains a representation limitation: there is no form for "all values except these tags".

7.3 Persisting incompleteness: no false-branch `instanceof`

R-Inst refines only the true branch; `else` branches keep the unnarrowed type. Sound (identity), incomplete relative to tsc.

7.4 Persisting incompleteness: multi-literal `in` keys

When the key expression's type is a union of literals, R-In keeps object members unchanged (still dropping primitives on the true branch); per-key case splits are not performed.

7.5 Known unsoundness: the `void`/`undefined` equality corner

The checker's `void` is a fiction over the runtime value `undefined` (Definition 2), and $\approx$ crosses the two tags (Definition 4) — which is exactly what the $\approx$-closure hypothesis of R-Eq guards. The corner is live:

```
function g(): void {}
function f(y: undefined): void {
  const x = g();
  if (x === y) {
    const n: never = x;  // accepted; branch taken at runtime
  }
}
```

A `void`-typed operand of `===` draws only a warning, so refinement runs; `y` is a variable, not the `undefined` literal, so the comparison takes the generic R-Eq rule, whose true branch installs $\mathsf{inter}(\texttt{void}, \texttt{undefined}) = \bot$ for `x` — the two units are distinct same-*model*-tag members and their intersection is empty, sound only under the closure hypothesis that fails here. At runtime `g()` returns `undefined`, `undefined === undefined` takes the branch, and `n : never` holds `undefined`. The dedicated R-EqUndef rule shows the required repair: treat the undef/void tags jointly wherever a comparison can relate them (assumption **L-SEQ**; issue report `deep-dives/issues/void-undefined-equality-never.md`).

8 Assumptions and Proof Obligations

Every axiom this document assumes, with its implementation anchor and a status from a fixed vocabulary: PROVED (established in this document or by inspection of the anchored code), ASSUMED (believed true, undischarged), OPEN (a stated obligation whose discharge is planned), and VIOLATED (a known counterexample exists in the implementation). A status may carry a note. Items whose discharge should be a pinned test say so in the note; wiring those tests is part of the maintenance plan (Appendix B). The obligations of Section 6 (Obligations 46–48) belong to this section's inventory; their current statuses (after the deep-dive series, `deep-dives/README.md`): Obligation 46 partially PROVED with named residues (deep dive D), Obligation 47 VIOLATED in two witnessed instances (see its status note), and Obligation 48 PROVED (deep dive E, with a documented exception).

Item	Statement	Status
L-TAG	Every runtime value carries exactly one tag consistent with its static tag.	PROVED
L-UNION	`union` over-approximates set union (all levels).	PROVED
L-INTER	When `inter` returns a non-$\bot$ result, that result contains the semantic intersection of its operands, provided every recursive call that returned $\bot$ had disjoint operands (L-INTER-BOT).	OPEN
L-INTER-BOT	A $\bot$ result from `inter` implies semantic disjointness.	VIOLATED
L-EXP	Expansion preserves or enlarges semantics (modes `None`/Covariant); residual instances keep theirs.	PROVED
L-MONO	Legal inference-state changes during refinement preserve membership in types computed earlier.	PROVED
L-SEQ	Type denotations used as equality RHSs are $\approx$-closed.	VIOLATED
L-ASSERT	`console.assert`(e) establishes e on the subsequent path.	VIOLATED

Continued on next page

Item	Statement	Status
L-SUB	`Clauses::isSub` acceptance implies semantic inclusion under the current state.	OPEN
A-CTOR-CALL	Class-constructor values carry CPresent construct and CFunction call modes, consistent with [[Call]]/[[Construct]].	PROVED
L-TC	Checked expressions inhabit their checked types.	ASSUMED
L-LIT	Literal spellings: (i) δ is a function; (ii) injectivity modulo $\approx$ after normalization; (iii) falsiness-exactness of the lexeme test.	VIOLATED
L-INST	$P_{\mathsf{inst}}(C) \subseteq [\![T_C]\!]$ for the instance type as instantiated at the narrowing site.	PROVED
L-ISARR	$\mathsf{isArrayV}(o)$ implies $o \in [\![\mathsf{ReadonlyArray}\langle\top\rangle]\!]$.	PROVED
L-LOCAL	Module-level bindings are never narrowed; function-level locals are, including captured ones.	PROVED
L-PURE	Evaluating a condition does not mutate the narrowed locals.	VIOLATED
L-CAPT	Assignments to captured annotated or initializer-inferred locals are checked against the binding's declared or widened inferred type.	PROVED
L-CAPT-EVOLVE	A captured write to an evolving binding (bare `let x = null/undefined`) is unchecked, and later reads keep the stale evolved type.	VIOLATED
L-STALE-READ	A closure body invoked later reads enclosing narrowings that are stale at invocation time.	VIOLATED
L-LOOP-CARRIED	A pre-loop narrowing of an annotated binding survives a back-edge write that invalidates it.	VIOLATED
L-CLONE	A type computed by `inter` under a cloned substitution over-approximates the intersection under the main one.	PROVED
L-MUT	Field writes are governed by set-types; get-only membership is sound for snapshot reads.	ASSUMED
L-MUT-ALIAS	A checked field or index write through one mutable view preserves the membership guarantees of every live alias of the same runtime object.	VIOLATED
L-GETSET	Get/set coherence of accessor fields.	OPEN

Details.

L-TAG (PROVED)

Every runtime value carries exactly one tag consistent with its static tag (the runtime counterpart of Lemmas A0/A1, which are proved only inside the model)

Anchor: ECMAScript language types / `typeof` (there is no SKJS code generator; checked programs run on a JS engine)

PROVED (deep dive A: realization map into ECMA-262 §6.1, modulo the environment assumption A-DDA)

L-UNION (PROVED)

`union` over-approximates set union (all levels)

Anchor: `TTypeUnion.sk`

PROVED (deep dive B — in fact exact at every representation layer for the membership semantics)

L-INTER (OPEN)

Non-bottom containment: when $\texttt{inter}(t_1, t_2) \neq \bot$, $[\![t_1]\!] \cap [\![t_2]\!] \subseteq [\![\texttt{inter}(t_1, t_2)]\!]$, conditional on L-INTER-BOT for recursive intersections used to build the result. The top-level $\bot$ case is also L-INTER-BOT; a consumer that concludes an inhabitant is *in* the implemented intersection needs both rows, since a $\bot$ result empties the conclusion

Anchor: `TTypeInter.sk`

PROVED (deep dive B) for every fresh structural intersection that returns non-$\bot$, conditional on L-INTER-BOT for recursive calls and using the canonical TVAR-SEM definition for the instance layer; OPEN for stale `seenInter` reuse under inference growth (C-FREEZE-CACHE)

L-INTER-BOT (VIOLATED)

A $\bot$ result from `inter` implies semantic disjointness — the property required by refinement rules that discard an impossible branch and by L-SUB's covariant variable read (O-GETCOV)

Anchor: `TTypeInter.sk`

VIOLATED (deep dive B) in two cases: the `TFun::inter` case — incomparable signatures treated as disjoint (issue report `tfun-inter-incomparable-not-disjoint.md`); and the num-kind beyond-precision spelling case — distinct spellings of one double kept distinct, their lexeme intersection a spurious $\bot$ (mediated by L-LIT; issue report `beyond-precision-numeric-literals.md`). The per-tag `STypes` assembly, within-precision numeric spellings, and decoded string spellings produce only semantic bottoms and are PROVED; stale `seenInter` reuse under inference growth remains the separate open cache obligation C-FREEZE-CACHE

L-EXP (PROVED)

Expansion preserves or enlarges semantics; residual instances keep theirs

Anchor: `TType::expand`

PROVED (deep dive C) for the modes refinement uses (`None`/Covariant) — and shown *false* in contravariant mode (unbound reads produce $\bot$), so the axiom is hereby scoped to those modes; all refinement clients verified to use only them. The containment proof consumes both L-INTER cases and is therefore conditional on L-INTER-BOT (VIOLATED) as well as on L-INTER's open cache-reuse case. Reusing an alias expansion is exact because alias normalization does not read existing variable bindings. The remaining conditions are recorded separately as E-NORM, E-ARR, E-CLASS-SYM, E-GUARD, E-EXPKEY, and E-PLACEHOLDER-ATOM; see deep dive C

L-MONO (PROVED)

Legal inference-state changes during refinement preserve membership in types computed earlier (Axiom 20)

Anchor: `Subst::get`, refinement sites

PROVED (deep dive E, given M1 — semantic monotonicity in variable envelopes — and L-UNION); `setConstraint` is a functional update, so there is no shared-state threat, and the range(Γ) restriction is droppable (see the axiom's remark)

L-SEQ (VIOLATED)

Type denotations used as equality RHSs are $\approx$-closed

Anchor: `evalVarEquality`

VIOLATED, witnessed end-to-end at the undefined/`void` corner (§7.5; issue report `void-undefined-equality-never.md`): `evalVarEquality`'s generic variable-RHS arm installs $\texttt{inter}(\texttt{void}, \texttt{undefined})$

`never` while ≈ relates the two units, and the `void`-operand check on `===` is a warning, so the corner is reachable — the branch runs with `undefined` bound at `never`. The undefined-*literal* arm filters the units jointly and is sound; literal/full primitives and objects remain ≈-closed by construction

L-ASSERT (VIOLATED)
The premise behind the `console.assert` refinement clause: the call must not return normally when e is false, and the callee must resolve to the global assertion function — neither condition holds

Anchor: `TypeCheckExpr.sk` (the `console.assert` match)

VIOLATED, witnessed end-to-end in both directions (issue draft `console-assert-narrowing.md`): a failed `console.assert` logs and continues, so the narrowed path runs with the condition false; and the match is by spelling alone, so a shadowing binding named `console` with an arbitrary `assert` method triggers the same refinement. tsc rejects both witnesses. Fix: remove the refinement or register a recognized assertion function that does not return when its condition is false

L-SUB (OPEN)
`Clauses::isSub` acceptance implies semantic inclusion under the current state

Anchor: `TTypeSubtyping.sk`

split (deep dive D): PROVED — dispatcher architecture, `tenv.seenSub` version-keying (corrected argument), coinductive reduction (via SEM-GFP), primitives, `STypes`, DNF covering, object skeleton, and field reads (conditional on L-SUB-IDXNAME at their numeric-named-field/dual-index corner); conditional — field *set* direction on L-MUT, functions on the E-SIG signature semantics; VIOLATED — `subst.seenSub` results recorded while candidate growth is suppressed (witnessed; issue report `freeze-window-stale-memo-caches.md`), and the two brands used by refinement but never validated by subtyping: a callable can enter a type marked `CAbsent` (F-CABSENT), and the never-compared array brand (F-BRAND)

A-CTOR-CALL (PROVED)
Class-constructor values are typed with construct mode CPresent and call mode CFunction (Definition Callability), so `typeof-"function"` refinement keeps them — the runtime facts [[Call]] and [[Construct]] are both represented

Anchor: `newStaticObject`; `narrowByTypeofFunction`

PROVED (deep dive A: constructor values built with both a construct signature and known callability, and the constructible-implies-[[Call]] invariant enforced at `TFields` construction); the remaining validation gap for the brand-like CAbsent claim is the separate row F-CABSENT

L-TC (ASSUMED)
Checked expressions inhabit their checked types (used for the RHS of R-EQ and field reads)

Anchor: `TypeCheckExpr.sk`

ASSUMED (axiom here; a full development would discharge it by mutual induction with a preservation theorem, of which the branch-soundness lemmas are the condition case)

L-LIT (VIOLATED)
Three claims about literal spellings: (i) δ is a function — definitional, used by R2; (ii) *injectivity modulo* ≈ after normalization — needed only for *exactness* claims; (iii) *falsiness-exactness*: the lexeme test `isFalsy` agrees with Definition 6 — needed by the residue lemma

Anchor: `normalizeJSNumber`/`normalizeJSBigInt`; `isFalsy`; string-escape decoding; enum fold outputs

(i) PROVED (definitional). For the *string* kind, (ii)/(iii) are PROVED: escape sequences are decoded at parse time (`JSStringEscape.sk`), so a literal's stored lexeme is its denoted string

— one spelling per value (ii) and the lexeme emptiness test agrees with falsiness (iii). For num/bigint, (ii)/(iii) are PROVED for *within-precision* literals: normalized to canonical form at parse time (`normalizeJSNumber`/`normalizeJSBigInt`), so on that range zero has one spelling per kind and injectivity holds. For *beyond-precision* numeric literals — those failing the round-trip `String(Number(s))` = s: 16–17-digit decimals, radix forms > 2^{53}, over/underflow exponent forms — (ii)/(iii) are VIOLATED: distinct spellings of one double are kept distinct, so `===` narrowing yields a spurious `never` (ii), and the normalizer can emit a zero-denoting lexeme (`1e-400`, underflow to +0) classified truthy (iii) — both witnessed end-to-end (issue report `beyond-precision-numeric-literals.md`). All three claims quantify over *all stored literals*, enum-folder-synthesized ones included (deep dive G proves the fold outputs canonical in form); whether a synthesized literal equals the runtime value of its source expression is the orthogonal row L-LIT-FOLD. All other results are unconditional

L-INST (PROVED)

$P_{\mathsf{inst}}(C) \subseteq [\![T_C]\!]$ for the instance type as instantiated at the narrowing site

Anchor: `evalInstanceof`

PROVED (deep dive A) modulo A-HASINST and A-PROTO (both ASSUMED) and, for intersected branches, L-INTER-BOT (VIOLATED): the containment step of the proof needs the intersection to be non-$\bot$ (L-INTER) — when it is $\bot$ the branch is dropped, which is sound only under the disjointness clause. Narrowing preserves the scrutinee's own type arguments — the instance view is built as `TFields` from the resolved class with the preservation probe run per branch — so the installed type is inhabited by exactly the values `instanceof` admits at those arguments (skolem interpretation: SKOLEM-SEM); no fresh unbound candidates are introduced at the narrowing site

L-ISARR (PROVED)

$\mathsf{isArrayV}(o)$ implies $o \in [\![\mathsf{ReadonlyArray}\langle\top\rangle]\!]$

Anchor: `evalIsArray`

PROVED (deep dive A: spec-level `IsArray` analysis + verification that the `__SKJS_isArray` brand is stripped into the `isArray` component, never a required field) modulo A-PROXY

L-LOCAL (PROVED)

Module-level bindings are never narrowed (they resolve through `defMap`, not the scope chain, so the primitive's early return fires). Function-level locals *are* narrowed — including locals of enclosing functions reached from inside a closure, since scope lookup walks the whole chain

Anchor: `setLocalFilterMap`; `TypeCheckScope.sk`

PROVED (by code inspection) for the module-level half; the captured-local half sharpens L-PURE; a pinned test is recommended. Narrowing module-level bindings is a tracked future enhancement (issue #585, blocked on immutability)

L-PURE (VIOLATED)

Evaluating a condition does not mutate the narrowed locals

Anchor: —

VIOLATED (empirically pinned): the checker narrows across a call that assigns a captured local and accepts the resulting unsound program (`if (x !== null && mut())` with `mut` setting `x = null`) — the same unsoundness tsc accepts. Tracked as issues #597 (side-effecting call between test and use) and #599 (captured local mutated via a lambda). The one-step scoping of the branch-soundness theorems (§1) is therefore a real boundary, not caution. Deep dive F specifies the fix: a layered invalidation policy (P1–P4) with a preservation lemma replacing the one-step scoping

L-CAPT (PROVED)

Assignments to captured annotated or initializer-inferred locals are checked against the binding's declared or widened inferred type (an implicit premise of L-TC for closure bodies); evolving bindings are the separate row L-CAPT-EVOLVE

Anchor: `Scope::set`; `TypeCheckExpr.sk` (lambda checking)

PROVED (deep dive F): assignments to captured annotated or initializer-inferred locals are routed through the binding's write path and checked against the declared or widened inferred type, in lambda bodies as at top level. The evolving-binding exemption is the separate violated row L-CAPT-EVOLVE; the read-side staleness of enclosing narrowings is the separate row L-STALE-READ

L-CAPT-EVOLVE (VIOLATED)

Evolving bindings are exempt from the captured-write check, and reads after the closure runs do not incorporate the written value — the part of the captured-write question that L-CAPT's proved statement excludes

Anchor: `Scope::set` (evolving exemption); `getSingleVar`

VIOLATED, witnessed end-to-end (deep dive F; issue report `evolving-capture-unchecked.md`): a closure writes `"s"` into a bare `let x = null` binding and a later read binds the string at type `null`; tsc accepts the same program (evolving `any` parity). Fix: track captured writes in the evolved union or check them against it at the write site

L-STALE-READ (VIOLATED)

A closure body is checked using an enclosing narrowing even when enclosing code can invalidate that narrowing before invoking the closure; #599 covers the opposite write-then-read direction

Anchor: `TypeCheckExpr.sk` (lambda checking); narrowed-scope capture

VIOLATED, witnessed end-to-end in both forms — a captured narrowed local and a narrowed field read inside a closure (deep dive F; issue report `stale-closure-reads-narrowing.md`); the fix is deep dive F's layered invalidation policy, shared with L-PURE

L-LOOP-CARRIED (VIOLATED)

Loop-carried narrowing staleness: `Scope::set`'s annotation arm checks a back-edge write only against the declared type, so a pre-loop narrowing of an annotated binding survives into iteration 2 after a write that falsifies it; un-annotated bindings are protected by the loop-write check

Anchor: `Scope::set` (annotation arm); loop merge

VIOLATED, witnessed end-to-end (deep dive F; issue report `loop-carried-narrowing-stale.md`): the narrowed read binds `null` at `string` on iteration 2; tsc rejects. Fix fix: deep dive F's P1 loop-entry invalidation layer

L-CLONE (PROVED)

A type computed by `inter` under a cloned substitution over-approximates the intersection when later read under the main one

Anchor: `evalVarEquality`

PROVED (deep dive E, fork-state bridging, modulo the fork-state forms of L-INTER and L-INTER-BOT); CL-1 is discharged by a non-escape audit: the `instanceof` preservation probe generates variables on a clone, but uses them only in subtype checks within the probe, so none reaches a type installed against the main substitution. The property should be guarded by the recommended CI assertion

L-MUT (ASSUMED)

Field writes are governed by `set`-types via subtyping; membership semantics (get-only) is sound for

the snapshot reads refinement performs

Anchor: `TField`

ASSUMED; full mutation model out of scope. The tracked instance is issue #598: field reassignment inside a refined branch does not re-widen the narrowed field, invalidating R-EqField/R-TruthyField results past the write; deep dive F specifies the re-widening rule (W-Field)

L-MUT-ALIAS (VIOLATED)

Cross-view mutation coherence, direct writes: a checked field or index write through one mutable view preserves the membership guarantees of every live alias of the same runtime object; call-mediated mutation is the separate row L-MUT-ALIAS-CALL

Anchor: `evalInstanceof`; indexed writes

VIOLATED (deep dive H) when an alias is created after field narrowing, for writes through either alias (issue report `narrowed-alias-stale-write.md`); types introduced directly by `instanceof` narrowing are coherent — the `Array` arm preserves the readonly index and the class arm preserves the scrutinee's type arguments — except when a readonly index covers a named field introduced by the class (V-MINT-ROIDX). Aliasing between declaration-derived types is coherent, conditional on L-SUB-FIELDS-SET

L-GETSET (OPEN)

Get/set coherence of accessor fields — the prerequisite for the sound accessor variant of deep dive F's W-Field re-widening rule

Anchor: `TField` (get/set split)

OPEN (deep dive F)

Series-introduced items. The deep-dive series introduces further named assumptions and obligations (interface facts its proofs consume, and residues its audits leave). They are registry entries like the rows above and carry the same status vocabulary:

Item	Statement	Status
L-LIT-FOLD	A literal synthesized by the enum constant folder denotes the member's runtime value.	VIOLATED
L-MUT-ALIAS-CALL	Calls that mutate a receiver (setter-style methods, `push`) through one view preserves the membership guarantees of every live alias.	OPEN
V-INTER-SET	Reachability of an incoherent live pair through the intersection's permissive `set`-union, independent of the already violated subtyping and narrowing sites.	OPEN
V-MINT-ROIDX	The `instanceof` class arm grants a write at a key the scrutinee covered only by a readonly index: the intersection's named field keeps a writable setter that survives `keepReadonlyFrom`.	VIOLATED
OBL-MONOTONE	Monotone Reads Under Growth: judgements validated under ρ survive later legal transitions.	VIOLATED
OBL-RESTORE	No Commitment on Failure: failing and exploratory paths restore all relevant state.	PROVED

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Item	Statement	Status
MR-1	No `subst.seenSub` memo recorded under `freezeCandidates` may outlive growth of an open candidate readable from its left side.	VIOLATED
C-FREEZE-CACHE	A stale `seenInter` entry can cause an unsound acceptance after candidate growth resumes without also reusing a stale `subst.seenSub` entry.	OPEN
CL-1	No fresh variable generated on a clone occurs in a type installed against the main substitution.	PROVED
M1	Semantic monotonicity of $[\![\cdot]\!]$ in variable envelopes (a property of the get-only, signature-opaque semantics).	ASSUMED
I-F	Fork-state form of L-INTER/L-INTER-BOT: non-bottom containment and bottom disjointness hold from a forked substitution state (consumed together by L-CLONE's proof).	ASSUMED
RB-1	Maintenance condition: `isSub` callers discarding errors are save/restore-wrapped or unreachable by uncommitted array literals.	OPEN
E-NORM	Named-type normalization is semantics-preserving (interface consumed by expansion).	ASSUMED
E-ARR	Delayed array-literal forms keep their semantics under expansion.	ASSUMED
E-CLASS-SYM	Class expansion either does not depend on existing variable bindings, or any bindings it reads cannot make cache reuse unsound.	OPEN
E-GUARD	Recursion-guard identity expansions are never cached into long-lived substitutions.	OPEN
SEM-GFP	The set-theoretic semantics is the greatest solution of the recursive membership equations.	PROVED
E-SIG	Behavioral call-soundness semantics for function signatures (required for function subtyping to be a soundness statement).	ASSUMED
A-HASH	`TType::compare` equality is trustworthy (XXHash-only fast paths).	ASSUMED
A-PROTO	No prototype surgery / Proxy GetPrototypeOf traps reach checked code (enforced by stub omissions).	ASSUMED
A-DDA	No [[IsHTMLDDA]] object (`document.all`) reaches checked code (vacuous on Node).	ASSUMED
A-PROXY	No Proxy exotic object reaches checked code.	ASSUMED
A-HASINST	No non-default `Symbol.hasInstance` governs a narrowed `instanceof`: narrowing is suppressed when the RHS declares one; dynamically installed hooks are assumed absent.	ASSUMED
TVAR-SEM	A type-variable occurrence denotes its substitution envelope intersected with its occurrence constraints; an unbound envelope is empty.	PROVED

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Item	Statement	Status
SKOLEM-SEM	A skolem in an installed `instanceof` view denotes the scrutinee object's actual hidden type argument, fixed at narrowing time and stable thereafter.	PROVED
SET-SEM	A view authorizes exactly the writes its resolved set-types accept; a readonly field authorizes nothing.	PROVED
L-SUB-FIELDS-SET	`isSub`'s field arms preserve write capability downward: acceptance implies each target field's set-type is below the source's, and readonly fields acquire no writers.	OPEN
L-SUB-CLOSED-ABSENCE	`isSub` admission cannot create a `Closed<T>` absence fact: acceptance requires the source to prove $\mathsf{get} = \bot$ at every key the target proves absent.	OPEN
O-GETCOV	Soundness of the covariant variable read in subtyping (inherits L-INTER-BOT's violated $\bot$-clause).	OPEN
O-GETCONTRA	Soundness of the contravariant variable read in subtyping.	OPEN
O-ARRAYLIT	Soundness of the array-literal decision procedure inside subtyping.	OPEN
F-CABSENT	The CAbsent brand-like claim accurately excludes callables: refinement drops CAbsent branches under `typeof "function"`, but subtyping's permissive (CUnknown, CAbsent) arm admits callables into branded types.	VIOLATED
F-BRAND	Subtyping never compares the array brand, though refinement narrows on it.	VIOLATED
F-THIS-IMPLICIT	Subtyping between two shape receivers, or two class receivers, with implicit `this` parameters accepts without comparing their receiver types.	ASSUMED
SETCONSTRAINTFUN	`TVar::setConstraint` is a functional update on a frozen value: constraints are occurrence-level, not shared state.	PROVED
E-TRANSPORT	Monotone-reads transport: an inclusion validated under ρ survives later legal transitions on the same timeline, except for the cache behavior recorded by MR-1.	PROVED
TENV-EXPANDINTER	The `tenv.expandInter` mode flag is path-scoped, and both intersection forms it selects (expanded and deferred `unexpandedInter`) satisfy the two-case L-INTER/L-INTER-BOT containment.	OPEN
L-SUB-IDXNAME	A numeric-named field is checked against both a number and a string index when the supertype has both (soundness of the width lemma's numeric-named case).	OPEN
A-INTERSEM	$[\![\mathsf{TInter}(\bar{t})]\!] \subseteq \bigcap [\![t]\!]$ — the instance-layer upper bound the dispatcher's fast paths assume.	ASSUMED
E-PLACEHOLDER-ATOM	An un-annotated ambient-class field reads as an opaque per-field atom; the atom's denotation must over-approximate the arbitrary runtime value.	OPEN

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Item	Statement	Status
E-EXPKEY	The expansion-cache key (`ExpansionKey`: unqualified name text + arguments) identifies one definition: same-named, same-argument aliases from different modules must not collide.	VIOLATED
ACCESSOR-METHOD	Accessor (`get`/`set`) members must be typed as fields, not methods; today a getter is a call signature, so `c.p()` type-checks and throws.	VIOLATED
L-SUB-TSINST	The reflexive `TSInst` fast path in `isSub` (same name, arguments equal after `peekRaw`) is sound.	VIOLATED

Details.

L-LIT-FOLD (VIOLATED)

A literal synthesized by the enum constant folder denotes the member's runtime value.

Anchor: `enumFold` (string-concat arms); `enumNumOrUnknown`

VIOLATED, witnessed end-to-end (deep dive G; issue report `enum-concat-imprecise-fold.md`): the string-concat arms concatenate operand *lexemes*, and a direct numeric-literal operand bypasses the numeric precision check that guards numeric fold results, so a beyond-precision spelling produces a member literal whose spelling is not the runtime string (`"x" + 1e-400` folds to `"x1e-400"`; the runtime value is `"x0"`). Numeric folds are exact within the checked range and produce no literal outside it — that half is PROVED. A prerequisite of the future preservation theorem's switch-coverage lemma (semantic case coverage consumes member-value faithfulness)

L-MUT-ALIAS-CALL (OPEN)

Cross-view mutation coherence for calls: a checked method call through one view preserves every live alias's membership; needs a receiver-effect strengthening of E-SIG, which as stated concerns behavioral signature semantics only

Anchor: method calls on narrowed receivers

OPEN (deep dive H): the live route is a mutating method call on an alias created after narrowing, but the signature semantics do not model effects on the receiver's fields

V-INTER-SET (OPEN)

`TField::inter` unions `set`-types; the open question is whether an incoherent live pair is reachable through it on a route not already covered by aliases created after narrowing or by direct narrowing

Anchor: `TField::inter` (set arm)

OPEN (deep dive H): the mechanism is established; every declared route probed is rejected by subtyping (the union widens the target's `set`, making admission harder), and refinement-installed intersections reduce to the sites already analyzed; ambient-declared intersection types enter through the stub trust boundary

V-MINT-ROIDX (VIOLATED)

When the scrutinee branch has a readonly covering index and no named field at key p while the class view contributes named p, `TField::interIndex` unions the setter types (the readonly $\bot$ is absorbed) and `keepReadonlyFrom` consults only named fields of the original — so the type introduced by narrowing authorizes a write the scrutinee type prohibited

Anchor: `TField::interIndex`; `keepReadonlyFrom`

VIOLATED by code-level derivation (found by the adversarial referee pass; no end-to-end run): the mechanism is verified in source, but three checked-code admission routes into a readonly-

covering-index view of a class instance were probed and all rejected — SKJS is stricter than tsc on those admissions — so the remaining candidate route is the ambient/stub trust boundary. Fix: make `keepReadonlyFrom` consult the original's covering index for named fields the intersection introduced

OBL-MONOTONE (VIOLATED)

Monotone Reads Under Growth: judgements validated under ρ survive later legal transitions.

Anchor: `Subst::get`; `TType::isSub`

VIOLATED in two witnessed instances — the freeze-window memo staleness MR-1 and the unsealed `peekRaw` read of the reflexive `TSInst` arm (L-SUB-TSINST, not memo-mediated); the general transport statement is the separate PROVED row E-TRANSPORT

OBL-RESTORE (PROVED)

No Commitment on Failure: failing and exploratory paths restore all relevant state.

Anchor: `SavedSubst`; save/restore sites

PROVED (deep dive E: 14-row save/restore audit; `SavedSubst` completeness; the intentional persistence of array-literal commitments is sound under RB-1)

MR-1 (VIOLATED)

No `subst.seenSub` memo recorded under `freezeCandidates` may outlive growth of an open candidate readable from its left side.

Anchor: `Subst::get` (freeze arm); `TType::isSub` memo

VIOLATED, witnessed end-to-end (order-dependent acceptance; deep dive E; issue report `freeze-window-stale-memo-caches.md`)

C-FREEZE-CACHE (OPEN)

A stale `seenInter` entry can cause an unsound acceptance after candidate growth resumes without also reusing a stale `subst.seenSub` entry.

Anchor: `seenInter`; `TType::inter`

OPEN for the isolated statement; mechanism demonstrated active (deep dive B); the MR-1 fix covers both memos

CL-1 (PROVED)

No fresh variable generated on a clone occurs in a type installed against the main substitution.

Anchor: `Subst::clone`, `Subst::gen`

discharged (deep dive E, non-escape audit): the one `gen*` site reached on a clone (the `instanceof` preservation probe) produces values consumed only as `isSub` right-hand sides on per-branch re-clones — no probe product is installed against the main substitution; every other `gen*` site runs on the expression-typing substitution, and the skolem class is hard-guarded at runtime

M1 (ASSUMED)

Semantic monotonicity of $[\![\cdot]\!]$ in variable envelopes (a property of the get-only, signature-opaque semantics).

Anchor: main paper semantics

ASSUMED (deep dive E; provable by structural induction in the main paper)

I-F (ASSUMED)

Fork-state form of L-INTER/L-INTER-BOT: non-bottom containment and bottom disjointness

hold from a forked substitution state (consumed together by L-CLONE's proof).

Anchor: `TTypeInter.sk`

ASSUMED (deep dives B and E); its two cases fold into L-INTER and L-INTER-BOT respectively

RB-1 (OPEN)

Maintenance condition: `isSub` callers discarding errors are save/restore-wrapped or unreachable by uncommitted array literals.

Anchor: `ArrayLitTyping`; `isSub` probe sites

OPEN maintenance condition (deep dive E); CI grep recommended

E-NORM (ASSUMED)

Named-type normalization is semantics-preserving (interface consumed by expansion).

Anchor: `TEnv::normalize`

ASSUMED (deep dive C)

E-ARR (ASSUMED)

Delayed array-literal forms keep their semantics under expansion.

Anchor: `TArrayLit`

ASSUMED (deep dive C)

E-CLASS-SYM (OPEN)

Class expansion either does not depend on existing variable bindings, or any bindings it reads cannot make cache reuse unsound.

Anchor: `newObject/addFields`

OPEN (deep dive C; proof and counterexample both outstanding)

E-GUARD (OPEN)

Recursion-guard identity expansions are never cached into long-lived substitutions.

Anchor: `seenExpansion`; recursion guards

OPEN (deep dive C); cache-write invariant + pinned test recommended

SEM-GFP (PROVED)

Greatest-fixpoint reading: $[\![\cdot]\!]$ is the greatest solution of the recursive membership equations.

Anchor: main paper object semantics

PROVED by definition in the main paper (every recursive occurrence in the membership conditions is positive, so the greatest solution exists); deep dive D supplies the stratified characterization used by the coinductive reduction

E-SIG (ASSUMED)

Behavioral call-soundness semantics for function signatures (required for function subtyping to be a soundness statement).

Anchor: main paper object semantics (extension)

ASSUMED (deep dive D; proposed extension of the main paper's object semantics)

A-HASH (ASSUMED)

`TType::compare` equality is trustworthy (XXHash-only fast paths).

Anchor: `TType` hashing

ASSUMED (deep dive D)

A-PROTO (ASSUMED)

No prototype surgery / Proxy GetPrototypeOf traps reach checked code (enforced by stub omis-

sions).

Anchor: import-stubs

ASSUMED (deep dive A; enforced for checked code by stub omissions)

A-DDA (ASSUMED)

No [[IsHTMLDDA]] object (`document.all`) reaches checked code (vacuous on Node).

Anchor: environment

ASSUMED (deep dive A)

A-PROXY (ASSUMED)

No Proxy exotic object reaches checked code.

Anchor: environment

ASSUMED (deep dive A)

A-HASINST (ASSUMED)

No non-default `Symbol.hasInstance` governs a narrowed `instanceof`: narrowing is suppressed when the RHS declares one; dynamically installed hooks are assumed absent.

Anchor: `rhsDeclaresHasInstance`

ASSUMED (deep dive A): the checker suppresses `instanceof` narrowing when the RHS class declares a custom `Symbol.hasInstance`, closing the declared case; the residue is a hook installed dynamically on an RHS whose declaration omits it — assumed unreachable for checked code, alongside A-PROTO's stub discipline

TVAR-SEM (PROVED)

Canonical `TVar` denotation: an occurrence denotes its substitution envelope intersected with its occurrence constraints; an unbound envelope is empty.

Anchor: `Subst::get/peek`

PROVED by definition in the main paper and matched to `Subst::get`: missing bindings read as bottom contravariantly and seal to top before a covariant expansion; occurrence constraints are intersected after either read

SKOLEM-SEM (PROVED)

Skolem interpretation: the clause parameters introduced by R-INST are interpreted by a per-view witness assignment — each skolem denotes the runtime object's actual hidden argument, chosen once when the view is installed. The assignment is stable because no later operation can bind a skolem: skolems are not inference variables, never enter the substitution, and clones cannot generate them

Anchor: `genSkolemTParam`; `makeSkolemTArgsOf`

PROVED: by definition in the main paper (the skolem interpretation definition) for the denotation, and by code inspection for stability — skolems live in clause var-sets, the substitution never binds them, and skolem generation in clones is rejected. Consumed by deep dive A's L-INST discharge (the existential step) and by deep dive H's per-skolem hidden-argument fact

SET-SEM (PROVED)

Write authorization: a view authorizes the write of v at key p iff v inhabits the resolved set-type of p (intersected named-field set-types, else the covering index's, resolved as at lvalue sites); readonly ($\mathsf{set} = \bot$) and keys with no writable field or index authorize nothing; call-mediated writes are E-SIG's

Anchor: `TFields::resolveFieldType` (`isLValue`)

PROVED by definition in the main paper and matched to the checker's write path: assignment checks the right-hand side against the resolved `set`-type by subtyping

L-SUB-FIELDS-SET (OPEN)

Write-capability preservation: field acceptance requires $f_2.\mathsf{set} \leq f_1.\mathsf{set}$ (contravariant) and the read-only check, so under SET-SEM an admitted view authorizes only writes the source view authorizes

Anchor: `TFields::isSub_`

OPEN (conditional): the syntactic checks are verified at every field arm (deep dive D, write-direction lemma); the SET-SEM containment they induce is conditional on the recursive `isSub` verdicts, exactly as the read direction is inside L-SUB

L-SUB-CLOSED-ABSENCE (OPEN)

Absence-fact preservation: against a closed target's never-valued index, a source with no covering index is rejected ($\top \leq \bot$), a source with one is accepted only if its `get` $\leq \bot$, and a source *naming* an extra field the target proves absent is rejected by the width lemma's extra-field check against the never-valued index — so an admitted view proves absent only keys the source already proved absent; the fact R-IN's true-branch pruning consumes

Anchor: `TFields::isSub_IndexPair`; `TFields::isSub_NamedFields` (extra-field arm); `closeObjectType`

OPEN (conditional): the syntactic checks are verified at every index-pair arm and at `isSub_NamedFields`'s extra-source-field-against-target-index arm (deep dive D, the absence corollary); the containment they induce is conditional on the recursive `isSub` verdicts, exactly as the read direction is inside L-SUB

O-GETCOV (OPEN)

Soundness of the covariant variable read in subtyping (inherits L-INTER-BOT's violated $\bot$-clause).

Anchor: `Subst::get` (covariant); `expandToClauses`

OPEN (deep dive D): the covariant variable read applies `Subst::get`'s constraint-intersection loop, whose soundness rests on the `L-INTER-BOT` bottoms; its live strata (incomparable signatures; beyond-precision num-spelling) leave the read unproven. On L-SUB's critical path

O-GETCONTRA (OPEN)

Soundness of the contravariant variable read in subtyping.

Anchor: `Subst::get` (contravariant)

OPEN (deep dive D)

O-ARRAYLIT (OPEN)

Soundness of the array-literal decision procedure inside subtyping.

Anchor: `ArrayLitTyping::subtyping`

OPEN (deep dive D; delegated)

F-CABSENT (VIOLATED)

The `CAbsent` brand-like claim accurately excludes callables: refinement drops `CAbsent` branches under `typeof "function"`, but subtyping's permissive (`CUnknown`, `CAbsent`) arm admits callables into branded types.

Anchor: `TFields::isSubCallability` (`CAbsent` arm); `narrowByTypeofFunction`

VIOLATED, witnessed end-to-end (deep dive D): a callable passes through `Function` $\to$ {} $\to$ `C` in two accepted subtyping steps, then is narrowed to `never` in the branch the runtime takes; issue report `cabsent-brand-laundering.md`. This has the same validation failure as F-BRAND: refinement relies on a brand-like claim that subtyping never validates

F-BRAND (VIOLATED)

Subtyping never compares the array brand, though refinement narrows on it.

Anchor: `TFields::isSub_`

VIOLATED by code-level derivation (deep dive D; composite witness sketch; no end-to-end run). The brand has a second consumer: the callability meet's array-brand $\bot$ arm (deep dive B) is semantic only under the brand's membership invariant, so a callable accepted into an array-branded view yields a spurious disjointness in a refinement rule that discards a branch

F-THIS-IMPLICIT (ASSUMED)

Subtyping between two shape receivers, or two class receivers, with implicit `this` parameters accepts without comparing their receiver types.

Anchor: `ThisParam` arms in `TFuns` subtyping

ASSUMED (deep dive D): these acceptances are justified only under an attachment semantics — a method invoked on the object it was accessed on — which E-SIG does not yet state; a detached object-*literal* method free-called with a `this`-reading body is accepted (probed), so the attachment premise is not checker-enforced on the literal route

SETCONSTRAINT-FUN (PROVED)

`TVar::setConstraint` is a functional update on a frozen value: constraints are occurrence-level, not shared state.

Anchor: `TVar::setConstraint`

PROVED (deep dive E, by code inspection: frozen class, functional update, both call sites use the return value)

E-TRANSPORT (PROVED)

Monotone-reads transport: an inclusion validated under ρ survives later legal transitions on the same timeline, except for the cache behavior recorded by MR-1.

Anchor: `Subst` transitions

PROVED (deep dive E, the monotone-reads theorem, given M1 and the MR/BR audit; subject to MR-1)

TENV-EXPANDINTER (OPEN)

The `tenv.expandInter` mode flag is path-scoped, and both intersection forms it selects (expanded and deferred `unexpandedInter`) satisfy the two-case L-INTER/L-INTER-BOT containment.

Anchor: `TEnv.expandInter`; `TInter::expand`

OPEN — found by external review: the flag is absent from deep dive E's unsaved-state inventory; path-scoping should follow the frozen-TEnv threading argument, and the deferred form needs deep dive B's containment coverage

L-SUB-IDXNAME (OPEN)

A numeric-named field is checked against both a number and a string index when the supertype has both (soundness of the width lemma's numeric-named case).

Anchor: `indexForFieldName`; `TFields::isSub_`

OPEN — found by external review: the width lemma checks a numeric-named sub field against only one super index (`indexForFieldName` returns a single index), while covering semantics needs both. Declarations whose number/string index value types are incompatible are rejected at the declaration itself (verified against the binary), shrinking but not closing the gap: the single-index consultation is still in code, unreached by any witness so far — a validator-completeness gap, not a witnessed unsoundness. Cited at deep dive D's width lemma as a named residual

A-INTERSEM (ASSUMED)

$[\![\mathsf{TInter}(\bar{t})]\!] \subseteq \bigcap [\![t]\!]$ — the instance-layer upper bound the dispatcher's fast paths assume.

Anchor: `TInter::expand`

ASSUMED (deep dive D; the instance-layer counterpart of deep dive C's `TInter` expansion — an upper bound on the denotation, distinct from L-EXP's over-approximation direction)

E-PLACEHOLDER-ATOM (OPEN)

An un-annotated ambient-class field reads as an opaque per-field atom; the atom's denotation must over-approximate the arbitrary runtime value.

Anchor: `instantiate` (missing-annotation placeholder)

OPEN (deep dive C): the placeholder is a fresh opaque atom, so the field neither head-expands to $\bot$ nor unifies with any other type — assignments out of it are rejected, which is conservative for the flows audited; outstanding is the atom's semantics under narrowing and equality (an arbitrary runtime value must inhabit it wherever membership is consulted)

E-EXPKEY (VIOLATED)

The expansion-cache key (`ExpansionKey`: unqualified name text + arguments) identifies one definition: same-named, same-argument aliases from different modules must not collide.

Anchor: `ExpansionKey`; `seenExpansion`

VIOLATED, witnessed end-to-end in both directions (deep dive C; issue report `expansion-key-collision.md`): two same-named, same-argument aliases from different modules share one `ExpansionKey`, so whichever expands first is served for both — an unsound acceptance one way (a `string`-typed call returns `undefined`) and an order-dependent false rejection the other, its diagnostic anchored in the wrong module; renaming either alias restores the correct verdicts. Refutes the source comment claiming the collision could not yet be triggered (upstream #889); fix direction: key by definition identity

ACCESSOR-METHOD (VIOLATED)

Accessor (`get`/`set`) members must be typed as fields, not methods; today a getter is a call signature, so `c.p()` type-checks and throws.

Anchor: `isGetSet` (parser-only); member typing

VIOLATED, witnessed end-to-end (SKJS-specific; tsc rejects with TS6234): the parser's `isGetSet` marker is never read by the checker, so a getter `get p(): T` is typed as a method `p(): T`. `c.p()` is accepted and throws at runtime (`c.p` is not a function). Issue report `getter-typed-as-method.md`. Adjacent to F's accessor caveat and L-GETSET, but a distinct (member-typing) hole

L-SUB-TSINST (VIOLATED)

The reflexive `TSInst` fast path in `isSub` (same name, arguments equal after `peekRaw`) is sound.

Anchor: `TType::isSub` (TSInst reflexive arm)

VIOLATED (deep dive D): the inclusion at the verdict state holds for unconstrained bindings, but the arm reads an unsealed `SVOpen` candidate via `peekRaw` without the covariant seal the slow path performs, so a later contravariant grow widens the binding and reads through aliases break — witnessed end-to-end (a value typed `number` holds a string; issue report `reflexive-tsinst-fastpath-unsealed-read.md`). For constrained variables the equality claim also fails as stated: the occurrence denotes $[\![t_2]\!]_\rho \cap \bigcap [\![C]\!]$, not $[\![t_2]\!]_\rho$

In summary, after the deep-dive series and its adversarial verification (series map in `deep-dives/README.md`; the authoritative per-item statuses are the tables above, from which this grouping is generated):

- *Proved*: L-TAG; L-UNION; fresh non-bottom intersections contain the semantic intersection, provided recursive bottom results are sound (L-INTER); L-EXP (mode-scoped; conditional on L-INTER/L-INTER-BOT and its named expansion obligations); inference growth preserves types computed earlier (L-MONO); the proved portions of L-SUB; A-CTOR-CALL; `instanceof` narrowing matches runtime instances (L-INST; subject to A-HASINST, A-PROTO, and L-INTER-BOT); L-ISARR; L-CAPT (annotated and initializer-inferred bindings); types computed on temporary inference-state clones remain sound in the main state (L-CLONE); OBL-RESTORE; SEM-GFP; TVAR-SEM; stable per-view interpretation of instanceof skolems (SKOLEM-SEM); SET-SEM.

- *Assumed*: checked expressions inhabit their checked types (L-TC); the full mutation model for field writes (L-MUT).

- *Open*: non-bottom containment after stale seenInter cache reuse (L-INTER; C-FREEZE-CACHE); the remaining conditional portions of L-SUB; getter/setter coherence (L-GETSET); whether stale `seenInter` reuse alone can cause unsound acceptance (C-FREEZE-CACHE); whether class expansion's binding reads permit exact cache reuse (E-CLASS-SYM); whether recursion-guarded expansions can enter long-lived caches (E-GUARD); covariant and contravariant variable reads and array-literal subtyping (O-GETCOV, O-GETCONTRA, O-ARRAYLIT); path scoping and containment for the intersection-expansion mode (TENV-EXPANDINTER); numeric-named fields checked against both numeric and string indexes (L-SUB-IDXNAME); the runtime meaning of the opaque type for unannotated ambient fields (E-PLACEHOLDER-ATOM).

- *Violated, with an executed witness* (each with a GitHub issue and an issue report under `deep-dives/issues/`): intersection returns $\bot$ for overlapping functions or numeric spellings (L-INTER-BOT); `undefined`/`void` equality can narrow a live value to `never` (L-SEQ); `console.assert` narrows even when a false condition can return normally or the receiver is shadowed (L-ASSERT); beyond-precision numeric spellings break equality and truthiness reasoning (L-LIT); incorrect string literal values from enum folding (L-LIT-FOLD); mutation can invalidate a narrowing during condition evaluation or inside a refined branch (L-PURE/L-MUT; #597–#599); unchecked captured writes to evolving bindings (L-CAPT-EVOLVE); stale closure reads of enclosing narrowings (L-STALE-READ); loop-carried narrowing staleness (L-LOOP-CARRIED); writes through one alias can invalidate another alias created after narrowing (L-MUT-ALIAS); later inference growth invalidates a cached result or an unsealed reflexive-`TSInst` read (OBL-MONOTONE); callables accepted into types marked `CAbsent` (F-CABSENT); the expansion-cache key collision (E-EXPKEY); accessor members typed as methods (ACCESSOR-METHOD); the reflexive `TSInst` fast path reads an unsealed candidate (L-SUB-TSINST).

- *Violated by code-level derivation* (no end-to-end run yet): instanceof narrowing can make a readonly indexed property writable (V-MINT-ROIDX); the never-compared array brand (F-BRAND).

The violations remain local to leaf assumptions: every fix restores its assumption without changing a theorem statement or proof in this document. Two design decisions determine several fixes and are tracked separately because a local patch cannot resolve them: unchecked brands — refinement relies on CAbsent and the array brand even though subtyping does not validate them (registry rows F-CABSENT, F-BRAND) — and the refinement-invalidation policy (deep dive F). The known unsound cases are enumerated and witnessed; their remaining fixes are future work.

None recorded.

A Implementation Correspondence

Pinned to `bafdcf7c`.

Model	Implementation
Tags (9 tags)	`STypeTag` (TType.sk)
unit, prim, object	`TPrimUnit`, `TPrimValue`, `TObject` (TType.sk)
literal payloads	`PrimLiterals`
object branches B, fields	`ArraySet<TFields>`, `TField`
Callability (CPresent/CFunction/CAbsent/CUnknown)	`TFields.call/new : Callability`
array brand	`TFields.isArray : ?Bool`
phantom fields	`FAPhantom`
STypes	`STypes` (tag-keyed map)
$\langle V \mid S \rangle$, Clauses, TType	`Clause`, `Clauses`, `TType`
refine^{g,d}	`setLocalFilterMap` + `TType::mapSTypes`
master list dispatch	`evalCond` (TypeCheckRefine.sk)
rmc	`TType::removeConstant`
tres$_b$, vres$_b$	`truthinessResidue`, `truthinessValueResidue`
scope merge $\sqcup$	`TCEnv::mergeScope`
ρ, transitions	`Subst`, `SVOpen`/`SVClosed`, `grow`/`get` (TTypeSubst.sk)
freeze	`Subst.freezeCandidates`
growth pre-pass	`growTVars` (TTypeSubtyping.sk)
validation	`Clauses::isSub`

B Maintenance

An advisory maintenance rule ("when the implementation adds a new refinement, add it to this list") does not survive contact with development. The replacement is enforced coupling; the parts marked *in place* are implemented by `docs/tools/docs.py` (`gen` / `check` / `build`):

1. **The registry** (*in place*). Statuses live only in `docs/ledger.yaml`; the tables of Section 8, their summary, and the deep dives' rows tables are generated from it (`docs/generated/`). A status change is a one-line registry edit; no document states another document's status.

2. **Consistency checks** (*in place*). The `check` command verifies: registry integrity (status vocabulary, anchor files exist, issue references resolve); every issue report is referenced and maps one-to-one to a distinct GitHub issue; no temporal/history markers in any document; no numeric cross-document references (deep dives must cite main-paper items by *name*); every document's date goes through the shared pin (`docs/shared/pin.tex`).

3. **Local builds** (*in place*). The `build` command compiles every document (two `pdflatex` passes), keeps each `.pdf` next to its source, fails on errors or unresolved references, and keeps auxiliary files in the git-ignored `docs/.texbuild/` for warm incremental recompilation.

4. **Anchor cross-check** (*pending*). Each dispatch arm of `evalCond` carries a comment `// paper: R-...` naming its rule; the check verifies both directions (every arm has a rule; every master-list rule has a live arm). Existence, not correctness — semantic drift within an arm still requires review, but silent additions and removals become failures.

5. **Witness repros** (*in place for issue witnesses*). Each executed unsoundness witness is a runnable `deep-dives/issues/repros/*.ts` with a header verdict; `docs.py repro` runs each against the pinned checker and asserts it, and `check` enforces that every repro is referenced by an issue report and vice versa. A repro that flips verdict when its fix lands fails, forcing a documentation update in the same change. Still *pending*: expect-error pins for the incompleteness worked examples in the test suite proper.

6. **Pinned revision.** `docs/shared/pin.tex` names the modeled commit once, for every document. A change that trips any check updates the pin as part of restoring it.

The review protocol for this corpus — scopes and checks for reviewing this paper, the deep dives, and the issue reports — lives in `docs/README.md`.